

Session 1 Medical Sciences-Biology

Prof Mohamed Salama

Professor at the American University in Cairo (AUC)
and senior fellow for the Global Brain Health Institute (GBHI)



Prof Mohamed Salama is a professor at the American University in Cairo (AUC) and senior fellow for the Global Brain Health Institute (GBHI)

He established the Egyptian Network for Neurodegenerative Diseases (ENND) in 2013 and the Egyptian Dementia Registry in 2023 as platforms to study diseases determinants. Currently, he is leading the NIH funded, Egyptian Survey of Healthy Aging (AL-SEHA) to study Egyptian seniors to identify determinants of healthy aging. Additionally, Funded by the Davos Alzheimer's Collaborative (DAC), Dr. Salama is leading the North African Dementia Registry (NADR) aiming at studying the various aspects of dementia in the region. He is also leading the Health and Wellbeing across Lifespan (HWLS) strategic research program within the African Populations Cohorts Consortium (APCC).

Risk and Resilience in Dementia: Towards a more proactive approach

The interplay of protective and risk factors influences healthy aging, particularly in high-inequality regions. This led to introducing the biological age gaps (BAGs) based on clocks, a novel framework for measuring the combined effects of protective/risk factors on delaying/accelerating aging, as well as their modulation by disparities and exposomes. Furthermore, scientific advances have unveiled several promising blood-based markers of health and aging. To have a useful model for studying dementia and brain health and more inclusive research is needed. Recently, Davos Alzheimer's Collaborative (DAC) has supported the initiation of an aging cohort in Egypt (modelled after HRS studies). The existing cohort is composed of 2000 subjects above the age of 50 who provided detailed history of exposure to risk factors, medical history and socio- economic status. Additionally, the cohort has been evaluated using recent digital biomarkers (speech and olfactory) aligned to clinical performance. DAC cohort study subjects provided blood samples, which will help generate blood based molecular (proteomics/epigenetics/metabolomics) data that explores

association of blood markers with Biological Clock calculated from cognitive, environmental and other available social determinants of health (SDOH)/clinical phenotyping data. This Egyptian cohort uniquely represents a high-inequality population with distinct lifestyle patterns, genetic background, and environmental exposures, critical for developing culturally and metabolically relevant aging algorithms. The existing infrastructure provides an exceptional opportunity to integrate multi-omics with longitudinal behavioral and cognitive outcomes. We are developing a multimodal Biological Clock by integrating peripheral blood biomarkers, functional/social/digital phenotyping, environmental data, and comprehensive clinical assessments. Identify novel aging targets and validate clock algorithms in the Egyptian population to quantify, monitor, and slow/reverse biological aging through identifying resilience determinants/model.

Prof. Dr. Nadia M. Hamdy

Prof. of Clinical Pharmaceutical Biochemistry and Molecular Biology, Faculty of Pharmacy,

Ain Shams University

Dept. Head (Feb. 2013-July 2015 & Sept. 2019- Sept. 2023)

GBD Senior Collaborator

<https://www.asu.edu.eg/staffPortal/en/staffProfile/1010>



Prof. Hamdy was a member in projects funded by NIH/AHA, European PRIMA, Erasmus+, ASRT-Egypt, STDF-Egypt, STDF-Germany, Higher Ministry of Education Egypt, “HME & Scientific Research”, ASU since 2004-till-now. Dr. Nadia has 140+ publications, 2 edited books and 14+ book chapters, with Scopus *h* index 52 and 40+ Masters’ and Ph.D. supervised students. Prof. Hamdy is one of the world 2% influencing Scientists, she earned Obada International Prize for the Life time achievement Scientists and 2 International BSI 2025 conference by Harvard University, honorary awards, Ain Shams University Appreciation award in Technological Sciences for the year 2024, the ASRT State Award Dr Shawky el Hadad award in cancer for the year 2023, International “Wendy Harvan” Top Honorary Award & Certificate in February 2023, “Paola Sassone-Corsi” Top Award in February 2021, and ASU first “Best Scientific Project Research Topic” 2021 & Honored by Women Of Egypt (WOE) 2020 to 2023 yearly. Dr. Nadia is an Ambassador of the EACR and Bentham Publisher. She is a member of the Biochemical Society & FEBS, IFCC, ISPE Mena Chapter, ESMO, AAAS, and more.

Dr. Hamdy is a fellow of the GBD, MHE Egypt, ASU, ASRT-Egypt. Prof. Nadia is a member of the editorial board of 10 Scopus (Q1 & Q2) journals and reviewer for hundreds of international and regional grants, awards, and scientific journals.

“(Epi)genetics Integration into Medicine for Advancing Precision via *Chem/Bio-informatics* Piloting/Validation”

Based on “disease-(epi)genetic profile” facilitates unraveling disease(s) mechanism(s), disease stratification via individuals’ (epi)genetic picture. State-of-the-Art *Study-for-Precision* ncRNAs-downstream protein or gene axis identified and predicted cellular:molecular processes via *in silico* and chem- or bio-informatics search and then verified experimentally and/or confirmed clinically, prompting “*Good Precision-Diagnosis or -Prognosis Practice*” (GPDP or GPPP). Finally, “*Good Precision Medicine Practice* (GPMP) via targeted therapies development or repurposing promising potential-to-repurpose drugs as the “*Old-is-Gold*” approach examining natural or chemically-modified small molecules drugs.

Prof. Dalia Osama Saleh

Pharmacology Department, Medical Research and Clinical Studies institute, National Research Centre, Egypt



Dr. Dalia Saleh obtained her BSc, MSc, and PhD degrees from the Faculty of Pharmacy, Cairo University, in 2002, 2009, and 2012, respectively. She is currently a Professor of Pharmacology at the Institute of Medical Research and Clinical Studies, National Research Centre (NRC), Giza, Egypt, located at 33 El-Bohouth Street, Dokki. She has also served as a course instructor at October University for Modern Sciences and Arts (MSA University) and Sadat City University. Dr. Saleh’s research interests encompass molecular pharmacology, biochemistry, cardiovascular diseases, in vitro pharmacology, and hepatic disorders, with a strong focus on mechanistic and translational research. She received the National Research Centre Scientific Pioneers Award in 2019 and the National Research Centre Scientific Encouragement Award in 2015, in recognition of her scientific contributions. She has published over 88 original research articles in high-impact international journals and holds a Scopus H-index of 21. Dr. Saleh has supervised more than nine Master’s and PhD theses in the field of pharmacological sciences. Dr. Saleh served as a Co-Principal Investigator for a project funded by the Science and Technology Center (STC), Academy of Scientific Research and Technology (ASRT) from 2019 to 2021. She has also contributed to several nationally and internationally funded research projects, including the Knowledge and Technology Alliances (EG-KTA) Program funded by ASRT (2017–2019), the Joint Research Grant Program (US–Egypt Fund/STDF) (2010–2014), and the Nanotoxicological and

Pharmacological Assessment of Some Nano-Gold Materials Used in Cancer Diagnosis and Therapy, a collaborative Egypt–USA project (2009–2014). In addition to her academic and research roles, Dr. Saleh is a Member of the Scientific Committee for Herbal Medicines at the Egyptian Drug Authority (EDA) and a Member of the National Committee for Women in Sciences, contributing to scientific policy, regulatory evaluation, and the advancement of women in science.

Natural Products as Multi-Target Therapeutics for Polycystic Ovary Syndrome

Polycystic ovarian syndrome (PCOS) is one of the most prevalent endocrine and metabolic disorders affecting women of reproductive age, characterized by hyperandrogenism, chronic anovulation, insulin resistance, obesity, and menstrual irregularities. The pathophysiology of PCOS is multifactorial and involves oxidative stress, low-grade chronic inflammation, mitochondrial dysfunction, and dysregulation of key metabolic and inflammatory signaling pathways. Increasing evidence implicates the aberrant activation of NF- κ B, NLRP3 inflammasome, and PI3K/AKT pathways, alongside impaired AMPK signaling and downregulation of the Nrf2-mediated antioxidant defense system, as central contributors to ovarian dysfunction and follicular arrest in PCOS. Recent preclinical and clinical studies highlight the therapeutic potential of natural products, including flavonoids, polyphenols, terpenoids, and alkaloids, in mitigating PCOS-associated reproductive and metabolic disturbances. These bioactive compounds exert pleiotropic actions by restoring hormonal balance, improving insulin sensitivity, regulating lipid metabolism, and preserving ovarian morphology. Mechanistically, natural products activate AMPK and Nrf2 signaling pathways, enhance endogenous antioxidant capacity, and suppress inflammatory cascades through inhibition of NF- κ B and NLRP3 inflammasome activation, ultimately reducing pro-inflammatory cytokine production. Collectively, accumulating evidence supports the role of natural products as promising adjunct or alternative therapeutic strategies for PCOS management. Targeting the crosstalk between oxidative stress, metabolic dysfunction, and inflammation through natural bioactive compounds may offer a safer, multi-targeted approach for improving ovarian function and metabolic health in women with PCOS.

Dr. Mawell Sameh, MD, CMD Head of Vaccines Egypt and Sudan



Biography: Dr. Mawell Sameh, a physician with two decades of clinical and pharmaceutical leadership experience, serves as Sanofi's Medical Affairs Head for Vaccines, where she has expanded access to life-saving medicines across Egypt and Sudan and earned over 10 Global & Regional Best Practice Awards, guided by her unwavering belief that no patient should be left behind.

Some careers follow a path. Dr. Mawell Sameh built one — from the wards of public hospitals to the forefront of vaccines leadership at one of the world's most innovative pharmaceutical companies.

Enhancing women's healthcare by vaccinations

Vaccination is a cornerstone of preventive healthcare and a key strategy for protecting women throughout their lives, particularly during pregnancy. Maternal immunization helps safeguard both mothers and their infants against serious vaccine-preventable diseases, improving health outcomes for families and communities. As innovation continues to expand the impact of preventive medicine, the commitment of healthcare partners such as Sanofi to developing vaccines that address the needs of pregnant women reflects the vital role of science in advancing maternal and newborn health. Together, research, innovation, and equitable access to vaccination can help build healthier futures for women and the generations they nurture.

Sherehan Khalil

Medical Operations Lead for Egypt within Sanofi's Africa Medical Operation department,
based in Cairo



Biography: Sherehan Khalil, MD is an internal medicine physician who transitioned to the pharmaceutical industry, bringing 8 years of clinical expertise and patient-centered perspectives to Sanofi. Based in Egypt, she has developed deep expertise in multi-market pharmaceutical environments and now leads with a focus on operational excellence and strategic impact in life sciences.

Diversity and Inclusion in clinical trials

Women remain significantly under-represented in clinical trials, yet they experience diseases differently and respond to treatments differently than men — a gap that perpetuates inequities in medical knowledge and care. Advancing diversity and inclusion in clinical research requires intentional action: recruiting more women participants, supporting women investigators as leaders in science, and designing trials that reflect the real-world needs of women across all populations.

Dr. Eman Abd El-Razek Metawea Abbas

Microbial Biotechnology Department, Biotechnology Research Institute, National Research Centre (NRC), Egypt



Dr. Eman Abd El-Razek Metawea Abbas is a Researcher at the Microbial Biotechnology Department, Biotechnology Research Institute, National Research Centre (NRC), Egypt. She specializes in medical biotechnology, molecular genetics, and translational cancer research, with a primary focus on hepatitis C virus, hepatocellular carcinoma (HCC), precision medicine, and molecular diagnostics. Her research aims to bridge the gap between molecular discoveries and clinical practice through the identification and validation of innovative biomarkers for hepatitis C virus-associated liver diseases and hepatocellular carcinoma. Her work encompasses host genetic polymorphisms, circulating cell-free DNA (cfDNA), liquid biopsy, genomic alterations, and non-invasive molecular biomarkers for disease susceptibility, early diagnosis, prognosis, survival prediction, and therapeutic response. Dr. Eman Abd El-Razek has authored numerous peer-reviewed publications in high-impact international journals covering host genetic biomarkers, liquid biopsy, and precision oncology. Her current research focuses on integrating genomic biomarkers with minimally invasive diagnostic approaches to support personalized medicine and improve clinical decision-making for liver cancer patients. In addition to her research activities, she serves as a reviewer for several international scientific journals and actively participates in multidisciplinary national and international research collaborations. She is committed to developing innovative molecular diagnostic strategies that enhance early cancer detection, prognostic assessment, and individualized treatment of liver cancer.

Host Genetic Biomarkers Associated with Disease Progression in HCV-Related Hepatocellular Carcinoma

Samar Samir Youssef¹, Eman Abd El Razek Abbas¹, Asmaa M. Elfiky², Sameh Seif³, Mohamed Mahmoud Nabeel⁴, Hend Ibrahim Shousha⁴, Ashraf Omar Abdelaziz⁴

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Background: Hepatocellular carcinoma (HCC) is a leading cause of cancer-related mortality worldwide. The marked biological heterogeneity of HCC highlights the need for reliable

molecular biomarkers to improve disease characterization and personalized patient management. **Aim:** This study aimed to investigate the clinical significance of PNPLA3 rs738409 and TM6SF2 rs58542926 polymorphisms and their association with disease progression and clinicopathological characteristics in patients with HCV-related HCC. **Methods:** Genomic DNA was extracted from peripheral blood samples of Egyptian patients with HCV-related HCC. Genotyping was performed using TaqMan allelic discrimination real-time PCR. Associations between genetic variants and clinicopathological parameters, tumor stage, radiological treatment response, and overall survival were analyzed. **Results:** TM6SF2 rs58542926 showed significant associations with advanced tumor stage and radiological treatment response, suggesting a role in disease progression. In contrast, PNPLA3 rs738409 demonstrated more limited clinical associations but contributed to the molecular characterization of HCC. Neither polymorphism was significantly associated with overall survival. **Conclusion:** Host genetic biomarkers may improve molecular risk stratification in HCV-related HCC. TM6SF2 rs58542926 demonstrated greater clinical relevance for disease progression, while combined evaluation of PNPLA3 and TM6SF2 may support future precision medicine strategies.

Session 2 Medical sciences and Biology

Prof. Reham Dawood

National Research Center, Egypt



Professor Reham Dawood is a Professor of Microbial Biotechnology at the National Research Center (NRC), Cairo, with more than 25 years of experience in molecular biology, virology, and translational biomedical research. She earned her Ph.D. in Molecular and Cellular Biology from the University of Jean Monnet, Saint-Étienne, France (2011), where her research focused on the development of broadly neutralizing monoclonal antibodies against HIV-1. Prof. Dawood has authored over 69 scientific publications with an H-index of 15 and more than 600 citations, reflecting her impact in the fields of viral genomics, immunogenetics, and biomarker discovery. Her research spans hepatitis C virus, COVID-19, cancer biomarkers, genome editing, and vaccine development, with several national and international grants led as Principal Investigator. She is also the Technical Supervisor of the DNA Fingerprinting Service at NRC and holds multiple gene bank accessions related to SARS-CoV-2. Her scientific excellence has been recognized by numerous prestigious awards, including the Prof. Mona Mohamed Rashoud Award in Biotechnology (2024) and the State

Encouragement Award in Health and Pharmaceutical Sciences (2020). Through her research, mentorship, and leadership, Prof. Reham Dawood embodies the vision of empowering scientific innovation and entrepreneurship, serving as an inspiring role model for women in science in Egypt and beyond.

Evaluation of lncRNA PVT1 rs13255292 Variant and Serum E-Cadherin levels in Breast Cancer

Breast cancer (BC) is the most frequent malignancy, and a prime cause of lethality related to cancer worldwide. Genetic research, particularly on lncRNA, shows prospects for BC management. PVT1 can activate many tumorigenic pathways. This contributes to angiogenesis and pathological progression. This study examined the association between the PVT1 rs13255292 variants and serum E-cadherin levels with BC risk, hormone receptor status, and tumor grade. Genotyping was performed on 120 blood samples (20 controls, 100 BC patients, 100 stratified by histological grade: G1=3, G2=74, G3=23) using TaqMan assays. Also, Patients were classified as luminal A (n=79) or non-luminal A (luminal B, HER2-enriched, TNBC) (n=21). Serum E-cadherin was evaluated by an ELISA kit. Findings revealed a statistically significant association between the PVT1 rs13255292 non-risk CC genotype and luminal A subtype patients, suggesting a potential protective effect. E-cadherin levels were significantly declined in BC patients compared to controls. Based on the histological grades, a notable reduction was detected in advanced G3 compared to G2. While serum E-cadherin showed promise as a non-invasive diagnostic biomarker. Also, the genotype-specific analysis indicated a trend toward higher E-cadherin expression in CC carriers' group, though without statistical significance. The current finding underscores that the CC genotype is associated with less aggressive luminal A tumors. It also reveals an inverse link between tumor grade and E-cadherin serum levels. These findings suggest that combining genetic screening of PVT1 variants with E-cadherin surveillance could enhance prognostic stratification in BC management. Further validation in larger cohorts is required to confirm clinical utility.

Prof. Dr. Menha Mahmoud Swellam

Head of High Throughput Molecular & Genetic Technology Lab, Center of Excellence for Advanced Sciences , Biochemistry Department Biotechnology Research Institute, National Research Centre



Menha Swellam is a Professor of Biochemistry at Biochemistry department- Biotechnology Research Institute, National Research Centre, Egypt. She is a member of editorial board and a peer reviewer of international journals. Supervised many MsCs and PhDs, a principal

investigator for national projects, member of the Egyptian permanent promotion council. Has published over 50 peer reviewed manuscripts in PubMed and Scopus cited journals (Scopus ID: <https://www.scopus.com/authid/detail.uri?authorId=56065210800>). She has been recognized in 2008 by Encouraging National Research Centre prize from National research Centre, and in 2009 by Encouraging State prize from the by the Academy of Scientific Research and Technology, then in 2017 a prize of National Centre for Research in Advanced Sciences. She is the Founder and the Head for High Throughput Molecular and Genetic Technology lab in the Centre of Excellences and Advanced Sciences, National Research Centre which is facilitated with next generation sequencers and microarray systems.

Clinical Impact of MiRNAs in Women Malignancies: What do we know so far?

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MicroRNAs (miRNAs) are endogenous, small non-coding RNAs which have an important function in regulating RNA stability and gene expression. Considerable proofs have demonstrated that miRNA expression is dysregulated in human cancer through several mechanisms, involving abnormal transcriptional control of miRNAs, amplification or deletion of miRNA genes, dysregulated epigenetic variations and limitations in the miRNA biogenesis machinery. MiRNAs may perform as either oncogenes or tumor suppressors through certain circumstances. The dysregulated miRNAs have been shown to influence the hallmarks of cancer, including escaping growth suppressors, supporting proliferative signaling, opposing cell death, triggering invasion and metastasis, and provoking angiogenesis. Cumulative numbers of studies have recognized miRNAs as potential molecular markers for human cancer diagnosis, prognosis and therapeutic targets, which desires more investigation and validation.

Ass.Prof. Ghada Salum

Department of Microbial Biotechnology, Biotechnology Research Institute, National Research Centre, Egypt



Dr. Ghada Maher Salum is an Associate Professor of Medical Biotechnology at the National Research Centre (NRC) in Egypt, specializing in virology and microbial biotechnology. She earned her Ph.D. in Microbiology (Virology) from Ain Shams University in 2016, focusing on the expression of interferon-alpha-related genes in Hepatitis C virus (HCV) induced liver diseases. Prior to this, she completed an M.Sc. at Cairo University in 2011, where she investigated the association of single nucleotide polymorphisms in the myxovirus resistance gene with response to interferon and ribavirin treatment in chronic hepatitis C infection. Her academic foundation was established with a B.Sc. in Microbiology & Chemistry from Zagazig University in 2004. Dr. Salum began her research career at the NRC in 2006, progressing through roles from research assistant to associate professor by 2022. She has contributed extensively to understanding viral-host interactions, immune response

modulation, and liver disease mechanisms, particularly in the context of hepatitis C and COVID-19. With 27 publications and a current H-index of 9, Dr. Salum's research outputs include original studies and reviews on hepatic fibrosis, liver injury related to COVID-19, and tumorigenesis-related microRNAs in breast cancer. She holds a patent registered with the Science Technology Development Fund (STDF) for a novel method that measures neutralizing antibodies in convalescent plasma to optimize donor selection for critically ill COVID-19 patients.

Genetic Dysregulation in Breast Cancer: Impact on Immune Surveillance and Infection Control

Ghada M. Salum¹, Mai Abd El Meguid¹, Basma E. Fotouh¹, & Reham M. Dawood¹

Department of Microbial Biotechnology, Biotechnology Research Institute, National Research Centre, EL Bohouth St. (Former El Tahrir St.), Dokki, P.O. 12622, Giza, Egypt.

Breast cancer (BC) is one of the main causes of cancer deaths in women worldwide, despite improvements in treatment. The understanding of genetic dysregulation in BC—specifically the alterations in tumor suppressors and oncogenes—provides crucial insights into how immune surveillance is compromised in these patients. This compromised immunity increases their susceptibility to infections, including those acquired in healthcare settings. Herein, we evaluated the expression levels of two circulatory miRNAs (miR-124-3p, and miR-155-5p) were assessed by using qRT-PCR in 75 BC patients (Luminal A: n = 24, Luminal B: n=12, HER2: n=28 and basal like: n = 5) and 20 healthy women. A computational analysis was carried out to identify the possible network between the two miRNAs and their related mRNAs involved in BC, which could help improve therapeutic options. Comparing to healthy controls, miR 124 3p levels were significantly lower in the serum of BC patients, however, miR 155 5p showed a significant elevation in BC patients specially in basal group. In-silico analysis suggested that CCND1 is an oncogene that is inhibited by miR-155-5p, was significantly overexpressed in luminal A & luminal B patients groups compared to controls. Moreover, CCND1 overexpression in basal like BC and HER2-positive subtypes do not achieve statistical significance. PRKD1 as a tumor suppressor was shown to be inhibited by miR-124-3p. PRKD1 showed a significant downregulation in the four BC subtype groups compared to controls. ROC curve analysis demonstrated that miR-124-3p possesses high diagnostic accuracy for distinguishing BC patients from controls. By integrating current findings into clinical risk assessments, infection control strategies can be personalized to identify high-risk BC patients, optimize preventive measures, and improve monitoring and targeted interventions. Thus, unveiling genetic dysregulation data can significantly enhance infection control effectiveness and patient outcomes in BC care.

Associate Prof. Ahmed Hassan Hassan Afifi

**Department of Pharmacognosy at the Pharmaceutical and Drug Industries Research
Institute, National Research Centre, Egypt**



Dr. Ahmed Hassan Afifi is an associate professor in the Department of Pharmacognosy at the Pharmaceutical and Drug Industries Research Institute, National Research Centre, Egypt. He holds a Ph.D. in Pharmaceutical Sciences (Pharmacognosy) from Kumamoto University, Japan, and Cairo University, Egypt, under the Channel System Program, along with an M.Sc. and B.Sc. in Pharmaceutical Sciences from Cairo University and Ain-Shams University, respectively. Dr. Afifi has extensive research experience spanning natural products chemistry, marine pharmacognosy, and drug discovery. He has held research and academic positions in Egypt, Saudi Arabia, and Japan, including a visiting researcher role at Kyushu University. His work has been published in leading journals such as the Journal of Natural Products, Tetrahedron Letters, RSC Advances, Scientific Reports, and Biomedical Chromatography. His research focuses on the isolation and characterization of bioactive natural compounds from plants and marine organisms, computer-assisted drug design and molecular modeling, and the discovery of novel anticancer, antiviral, and neuroprotective agents. He possesses strong expertise in chromatographic separation, spectral analysis, and metabolic profiling using LC-MS and GC-MS. Dr. Afifi has participated in numerous national and international research projects and has presented his work at conferences in Egypt and Japan, including meetings of the Japanese Society of Pharmacognosy and the Asian Natural Product Conference. He continues to contribute to advancing drug discovery through interdisciplinary, technology-driven approaches to natural product research.

Marine Environment as a Potential Source for Small Molecule Targeted Anti-Cancer Agents

Abstract: Cancer remains one of the leading causes of death globally, responsible for approximately 10 million deaths in 2022, and is projected to claim 18 million lives by 2050. While conventional approaches such as chemotherapy and radiation therapy have long been employed for cancer treatment, their non-specific nature often results in severe side effects and the development of resistance in cancer cells. Targeted cancer therapy has emerged as a new generation of cancer treatment, providing a more precise approach by interfering with certain molecular targets unique to cancer cells while sparing healthy tissues. Marine environments, known for their exceptional biodiversity, have yielded a rich array of bioactive compounds with unique structures and potent anticancer properties. This presentation highlights the role of marine-derived small molecules in targeted cancer therapy, particularly those modulating key cancer specific molecular targets such as protein kinases, ubiquitin-proteasome system, epigenetics, and other molecular drivers of cancer. With several marine-derived agents already in clinical use, the continued exploration of these natural products holds great promise for developing novel, more effective, and less toxic treatments in the fight against cancer.

Ass. prof . Mona Mohammad Agwa

Department of Chemistry of Natural and Microbial Products, Pharmaceutical and Drug Industries Research Institute, National Research Centre



Mona M. Agwa, Associate Professor, Department of Chemistry of Natural and Microbial Products, Pharmaceutical and Drug Industries Research Institute, National Research Centre, , Egypt . Ph.D, from Institute of Graduate Studies and Research, Alexandria University in 2017. Master degree from Institute of Graduate Studies and Research, Alexandria University in 2011, and Bachelor's degree from Faculty of Pharmacy, Al Azhar University, Egypt in 2002. Has contributed to 46 scientific papers published in international journals (indexed in Scopus/Web of Science) (H-index 20) and authored four book chapters for Taylor & Francis/CRC Press and Elsevier on functional chemicals and advanced pharmaceuticals. Additionally, acted as a peer reviewer for leading international publishers including Elsevier, Springer Nature, Bentham, and Dove Press ”having reviewed over 20 manuscripts. Also, co-supervised PhD and Master theses and participated in more than 5 research projects focused on the development of pharmaceutical dosage forms and drug delivery systems. Additionally, supervised and judged many graduation projects for undergraduate students at MSA University. She participated in teaching training courses on targeted drug delivery, different routes of drug administration, overcoming hazards of chemotherapy, and emerging role of chemotherapy/ phototherapy based nanomedicine as a non-invasive therapeutic approaches for tumor treatment.

Active tumor targeting exploiting overexpressed receptors: Emphasize towards harnessing cellular and subcellular targeting ligands.

Nano-delivery systems for actively targeting drugs against diseased tissues like tumors is a prime goal of nanomedicine, which ameliorates the therapeutic delivery to the aimed tissue while evading the off-target toxicity issue and the necessity for dosing several doses. An Active targeting approach is achieved via decorating the nanoparticle surface with various targeting ligands, harnessing chemical conjugation. At the cellular level, these targeting ligands enable selective tumor homing for the nanocarriers via recognizing the overexpressed cellular receptors or transporters, thereby guaranteeing excellent drug efficacy in the aimed tissue with negligible toxicity. In addition to cell-specific targeting, these ligands can also assist active subcellular therapeutic targeting to particular intracellular organelles, involving lysosomes nuclei, endoplasmic reticulum, and mitochondria, where multiple pathological processes originate. Integrating cellular and subcellular targeting creates a hierarchical delivery system that maximizes drug availability at the site of action, minimizes systemic toxicity, and holds strong promise for treating complex diseases.

Keywords: Active targeting, cellular and subcellular active target ligand, chemotherapy, Overexpressed cellular and subcellular receptors.

Dr. Basma Elsayed Fotouh

National Research Centre, Egypt

I am a Researcher at National Research Centre. I have experience in molecular biology and medical biochemistry.



Impact of KRAS gene expression on ovarian cancer development and progression

Ovarian cancer (OC) is a leading cause of cancer deaths in women. Comprehensive molecular studies are required to understand OC pathogenesis. KRAS and NOXA genes are involved in tumorigenesis and disease progression. KRAS promotes tumor growth, while NOXA triggers the apoptotic signaling pathway. Fifty-six ovarian cancer patients and twenty healthy controls were enrolled in the study. Gene expression profiling was performed utilizing the qRT-PCR assay. Then, the proteomic levels of KRAS and NOXA genes were assessed by the ELISA technique. The results showed that KRAS gene expression was significantly increased in OC patients compared to controls ($p < 0.0001$). Additionally, KRAS overexpression was correlated significantly with more aggressive characteristics, including advanced stage, positive lymph invasion, and metastatic status ($p = 0.04$, 0.05 , and 0.02 , respectively). In contrast, NOXA expression was downregulated in OC patients compared to controls ($p = 0.001$), and this reduction was more pronounced in patients with more aggressive characteristics ($p = 0.001$, 0.01 , 0.0008 , 0.02 , respectively). At the transcriptomic level, KRAS concentration was higher among the patient group ($p = 0.03$) and correlated with aggressive tumor features ($p = < 0.0001$, 0.001 , and 0.03 , respectively). Interestingly, no significant changes were detected in NOXA protein levels concerning ovarian cancer development and progression. These findings were further confirmed by receiver operating characteristic (ROC) curve analysis, validating the genetic and transcriptomic results. The differential expression of KRAS and NOXA genes holds promise as potential biomarkers for ovarian cancer development and progression. Specifically, KRAS transcriptomic levels serve as a reliable discriminator of ovarian cancer progression. In contrast, NOXA protein expression warrants further investigation to elucidate its role in the progression and pathophysiology of ovarian cancer.

Dr. Tânia dos Reis

Instituto de Higiene e Medicina Tropical and Universidade of Cape Verd, Cape Verd



I am a Biotechnology graduate, Bioinformatics specialist, and MSc in Genetics and Molecular Biology, currently pursuing a PhD in Tropical Diseases and Global Health. My academic trajectory is rooted in interdisciplinary research, global health equity, and scientific innovation, with a strong commitment to leadership, capacity building, and social impact across Africa and the Global South. I have extensive experience in scientific research, having worked in multiple laboratories and participated in interdisciplinary projects addressing health challenges relevant to low and middle income countries. I have contributed to the scientific community through conference presentations and the organization of scientific symposia and workshops that promote knowledge exchange beyond geographical borders. I also bring over five years of teaching experience in biomedical sciences, alongside academic leadership experience as a course coordinator and founder of a health sciences research group. My doctoral research focuses on the respiratory microbiome in children, investigating differences among those with asthma, allergic rhinitis, and healthy individuals. This study employs innovative laboratory techniques, including advanced molecular biology approaches and high throughput sequencing, integrated with state of the art bioinformatics and computational analyses. By combining microbial diversity profiling, environmental exposure data, and clinical information, my research aims to advance understanding of microbiome–environment interactions and their role in allergic disease development, contributing to precision and preventive health strategies adaptable to resource limited settings. Beyond academia, I am committed to science communication, mentorship, and gender equity advocacy. Through the WFWS and WISWB Science Without Borders platform, I seek to strengthen international collaborations and amplify the voices of women scientists from Africa and the Global South.

The title of your presentation: Nasal microbiota and its relationship with asthma and other atopic conditions in children from Cabo Verde

The role of the human microbiota in the pathogenesis of allergic diseases is well established, however, the specific contribution of the nasal microbiota remains less understood. Emerging evidence suggests that alterations in the composition and diversity of the respiratory microbiota may contribute to the development of asthma and other atopic conditions in susceptible individuals. Asthma is the most prevalent non-communicable disease in children, with a major burden in low- and middle-income countries. In Cabo Verde, a population-based study conducted involving 1,045 children reported a current asthma prevalence of 10.5%, exceeding the global average of 9.1%. A case-cohort study is currently in progress and aims to investigate the relationship between alterations in the nasal microbiota, atopic conditions, and the risk of upper respiratory tract colonization by pathogenic microorganisms among children living on Santiago Island, Cabo Verde. Here we report the preliminary characterization of the nasal microbiota of children from that study, based on analyses of two batches of sequenced samples in a total of 142 samples, including 67 cases (current or

previous respiratory atopy) and 75 control samples (no atopic condition). Nasal swab samples collected from children were subjected to DNA extraction followed by amplification of the V3–V4 hypervariable regions of the 16S rDNA gene in the Illumina NovaSeq PE250 platform. After quality control analysis, the reads were clustered into OTUs by vsearch. Taxonomic assignments were based on the SILVA reference database. In the first batch of samples, 70k paired-end reads per sample were obtained. Preliminary taxonomy analyses revealed more than 200 families of bacteria and the presence of some bacterial species commonly associated with respiratory diseases in children (*Staphylococcus aureus*, *Haemophilus influenzae*). Alpha-diversity indices and beta-diversity will be evaluated to assess differences between case and control groups. Differentially abundant taxa will be identified through linear discriminant analysis. The study is expected to provide deeper understanding of the relationship between airway microbiota composition and asthma and other atopic phenotypes. The findings may support the development of personalized therapeutic strategies for pediatric allergic diseases.

Dr. Walaa Abdallah Gad

Veterinary Research Institute, National Research Centre (NRC), Egypt



I'm Dr. Walaa Abdallah Gad, a researcher at the Veterinary Research Institute, National Research Centre (NRC), Egypt, specializing in infectious diseases, molecular diagnostics, biotechnology, immunology, and bioinformatics. My research focuses on developing innovative diagnostic and therapeutic strategies for veterinary infectious diseases, antimicrobial resistance, molecular epidemiology, and One Health applications. I have extensive experience in molecular biology techniques, including PCR, real-time PCR, sequencing, gene expression analysis, immunological assays, and bioinformatics. During the past year, I have significantly expanded my scientific contributions through research, publishing, and academic editing. I currently serve as an Editor of an international scientific book and have authored eight book chapters, three review articles, and five original research articles, reflecting my commitment to advancing knowledge in veterinary medicine and molecular biosciences. My research interests include rapid molecular diagnostics, CRISPR-based technologies, isothermal amplification techniques, epigenetics, precision veterinary medicine, and artificial intelligence applications in biomedical research. Beyond research, I actively organize and deliver scientific workshops and professional training programs on molecular diagnostics, gene expression analysis, bioinformatics, and advanced laboratory techniques for researchers and veterinary professionals. I also serve as a peer reviewer for international scientific journals, contributing to the advancement of high-quality scientific publishing. My long-term goal is to bridge laboratory innovation with practical veterinary

applications by developing affordable, rapid, and sustainable diagnostic solutions that improve animal health, reduce antimicrobial resistance, and support global One Health initiatives through interdisciplinary collaboration and translational research.

Epigenetics and Gene Therapy Technologies for Precision Veterinary Medicine

Mastitis remains one of the most prevalent and economically significant diseases affecting dairy livestock worldwide, compromising animal welfare, milk quality, and farm profitability. Although antimicrobial therapy has long been the cornerstone of mastitis management, the emergence of antimicrobial resistance, recurrent infections, and treatment failures has emphasized the need for innovative molecular-based interventions. This review explores the rapidly evolving roles of epigenetics and gene therapy as promising strategies for the prevention, diagnosis, and treatment of mastitis in veterinary medicine. We discuss the molecular mechanisms underlying host pathogen interactions, mammary gland immunity, and inflammatory responses, highlighting how DNA methylation, histone modifications, and non-coding RNAs regulate susceptibility and resistance to infection. The review further examines the influence of the mammary microbiome and its interaction with epigenetic pathways in maintaining udder health or promoting disease. Recent advances in epigenetic modulators, including DNA methyltransferase, histone deacetylase, and bromodomain inhibitors, are evaluated for their therapeutic potential and limitations in food-producing animals. In addition, current gene therapy technologies, including CRISPR/Cas systems, viral and non-viral vectors, antimicrobial peptide delivery, and regulatory RNA-based approaches, are critically assessed with respect to efficacy, safety, and tissue-specific delivery challenges. The integration of epigenetic editing, multi-omics technologies, artificial intelligence, and bioinformatics is presented as a transformative framework for precision veterinary medicine, enabling biomarker discovery and individualized disease management. Finally, the review addresses regulatory, ethical, and practical considerations associated with implementing these advanced technologies in livestock production. Collectively, this review highlights how combining epigenetic insights with gene therapy may provide sustainable, antibiotic-sparing solutions for mastitis control while advancing the future of precision livestock health management.

Dr. Amal Abdel-Bassir Draz (Short talk)

**Phytochemistry and plant systematics department , Pharmaceutical and Drug
Industries Research Institute**



Dr. Amal A. Draz is a researcher at the National Research Centre (NRC), Egypt, specializing in plant chemotaxonomy. She earned her Ph.D. in Plant Taxonomy and Flora from the Faculty of Science, Cairo University, in 2025, following a Master's degree in the same field (2021). She also holds a Master degree in Microbiology (2014) from Benha University and graduated with a B.Sc. in Botany and Chemistry with Excellent with Honors (86.77%) in 2009. Dr. Amal Draz joined the National Research Centre as a Research Assistant (2017–2021), was promoted to Assistant Researcher (2022–2025), and has served as a Researcher since 2026. She also held the position of ISO Quality Coordinator in the Phytochemistry and Plant Systematic Department (2024–2026). Earlier in her career, she worked as a Chemist in the field of Safety and Occupational Health (2012–2017). Her research interests focus on plant taxonomy, flora, systematics, and related biological sciences. She has published 11 research papers in international peer-reviewed journals and received the National Research Centre Best Master Thesis Award in 2021. Dr. Amal Draz has completed specialized training in artificial intelligence applications in research, and young researcher capacity building. She is also a certified TOEFL-ITP holder (score: 573) and has qualifications in digital transformation, reflecting her commitment to continuous professional development.

From authentication to Bioactivity: Metabolite profiling, DNA Barcoding, and Antimicrobial Evaluation of *Zygophyllum scabrum* Christenh. & Byng.

The growing threat of antimicrobial resistance highlights the urgent need for novel plant-derived therapeutics supported by reliable authentication methods. This study integrates DNA barcoding, metabolite profiling, and antimicrobial evaluation to characterize *Zygophyllum scabrum* and assess its pharmaceutical potential. Species authentication was achieved using the *rbcl* DNA barcode, resulting in a novel registered NCBI GenBank accession (PQ403596), representing the first reported *rbcl* sequence for this species. Comprehensive phytochemical profiling of 70% methanolic extract of the plant by LC-ESI-MS/MS tentatively identified 97 metabolites, including flavonoids, phenolic acids, saponins, coumarins, and fatty acids. On the other side, in a first evaluation of the antimicrobial activity of this plant, the methanolic extract demonstrated notable antimicrobial activity comparable with that of the standard antibiotic disc, as determined by the agar well diffusion assay, against *Helicobacter pylori* (14.0 $\pm$ 0.05 mm), *Bacillus cereus* (13.0 $\pm$ 0.08 mm), and *Staphylococcus aureus* (11.0 $\pm$ 0.09 mm). The minimum inhibitory concentrations (MICs), determined using the broth microdilution method, ranged from 1875 to 3750 μ g/ 300 μ L. In contrast, no antimicrobial activity was observed against *Escherichia coli* or *Candida albicans*. These findings demonstrate that *Z. scabrum* represents a promising medicinal plant with significant phytochemical diversity and antibacterial potential. The integration of molecular authentication with metabolomic profiling provides valuable data for the pharmaceutical quality control of medicinal plants and their potential pharmaceutical applications.

Ass. Prof. Ola Mohamed Mousa

Faculty of Nursing, Minia University, Egypt



Dr. Ola Mousa is currently an assistant professor at the faculty of nursing, Minia University. She worked for more than 10 years (from January 2013 to September 2023) as an associate professor at King Faisal University, College of Applied Medical Sciences- Nursing Department, Saudi Arabia. She started her career in Minia University, Egypt and taught there for 13 years from 2000 to 2013. She earned her master's from Ain Shams University, Egypt, and PhD degree from Assiut University, Egypt with a specialization in Maternal and Women's Health Nursing. Her expertise in healthcare professions extends over 23 years, both nationally and internationally. Her area of interest in research is Maternity Nursing, Ethics, Women and Adolescent Health, and mothers of children with special needs. She participated in organizing many scientific days and conferences, as a resource speaker. In addition to publishing more than 55 papers in significant journals, including those indexed in Web of Science and Scopus, she has presented research articles at numerous national and international conferences. She served as an editorial board member and reviewer in many journals. In addition to being a thesis advisor, she is a research guide for MSN students. Over the past seven years, she has been responsible for community engagement and participation in activities with special needs pupils and caregivers. A distinguished faculty award was recently received (2020) from King Faisal University in Saudi Arabia. Presently, she is teaching at Minia University, Faculty of Nursing, the chairperson of the Quality Assurance Department Committee, and Faculty Development committee.

From Awareness to Action: Exploring Risk Perception and Behavioral Patterns of Antibiotic Misuse Among Minia University Students

Self-medication, defined as the use of pharmaceutical products without professional prescription to manage self-diagnosed or recurrent symptoms, has emerged as a major contributor to antibiotic misuse. This practice is particularly prevalent among adolescents and university students, raising concerns due to its irrational nature and long-term public health implications. Factors such as limited awareness, permissive attitudes, and unregulated access to antibiotics at pharmacies exacerbate the problem. The consequences include not only the proliferation of resistant bacterial strains but also adverse drug reactions and increased economic burden on national health systems. In the United States alone, antibiotic-resistant infections are estimated to cost \$35 billion annually and account for over 8 million additional hospital days. To assess the level of awareness, attitudes, and behavioral practices regarding antibiotic misuse among medical and non-medical students at Minia University, and to identify the key factors influencing their knowledge and decision-making. This study employed a descriptive cross-sectional design to assess the knowledge, attitudes, and practices (KAP) related to antibiotic misuse among medical and non-medical undergraduate students at Minia University. A sample size of [400] students was calculated based on previous prevalence estimates of antibiotic misuse among university students, with a confidence level of 95% and a margin of error of 5%. A stratified random sampling method was used to ensure proportional representation from both medical and non-medical faculties.

Data Collection Tool

- Demographics: age, gender, faculty, year of study.
 - Knowledge: 10 items assessing understanding of antibiotic use and resistance.
 - Attitudes: 6 items exploring perceptions and concerns about misuse.
- Practices: 8 items evaluating self-reported behaviors related to antibiotic use.

Session 3 Medical sciences and Biology:

Prof Nesrine Hegazi

**Phytochemistry and Plant Systematics Department, Pharmaceutical Industries Institute,
National Research Centre**



I am a graduate of the Faculty of Pharmacy, Cairo University, Egypt (2003). MSc (2008) and PhD (2014) degrees from Ain Shams University, Egypt, in Pharmaceutical Sciences. Professor at the National Research Centre in the Department of Phytochemistry and Plant

Systematics since 21/4/2026. My research interests are natural products chemistry, with a special emphasis on recent advances in metabolomics tools and their application.

From Sea to Table: The Power of Metabolomics in Marine Food Science and Facing Challenges

Abstract

Marine food science has gained increasing attention owing to its potential to offer valuable bioactive compounds beneficial for human health aside from nutritive value. Metabolomics, a cutting-edge analytical technology aims to provide a comprehensive insight into the chemical composition of marine food systems, ensuring quality and safety. This review explores the integration of metabolomics in marine food science, emphasizing its ability to profile metabolites for quality control, authentication, and safety determination in seafood products. Metabolomics can aid in identifying geographical origin, detecting contaminants, and monitoring freshness levels in seafood. Additionally, it plays a critical role in advancing aquaculture by monitoring the health and nutritional quality of farmed fish species. The use of advanced spectroscopy exemplified by nuclear magnetic resonance (NMR) and mass spectrometry (MS) allows for the discovery of novel bioactive compounds with therapeutic potential, enhancing the nutritional value of seafood. Despite the challenges in data interpretation due to the complexity of marine metabolites and their origin, the combination of metabolomics with other "omics" technologies held promise to revolutionize our understanding of marine ecosystems. Further research is needed to overcome current analytical challenges and translate these findings into practical applications for the food industry.

Keywords: Marine food; metabolomics; aquaculture; authentication

Ass.Prof. Mona Abd El-Mouty Moustafa Raslan

**Department of Pharmacognosy at the Pharmaceutical and Drug Industries Research
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Dr Mona Raslan is an Associate Research Professor in the Department of Pharmacognosy at the Pharmaceutical and Drug Industries Research Institute, National Research Centre. Dr Raslan holds a Ph.D. in Pharmaceutical Sciences from Cairo University. She has extensive

research experience in natural products chemistry and pharmacognosy. Dr Raslan has published numerous research articles in prestigious journals such as Phytochemistry Letters, Egyptian Journal of Chemistry, New Journal of Chemistry, and Chemistry and Biodiversity. Dr Raslan's research focuses on various aspects of natural products research, including the isolation and characterization of natural compounds, evaluation of their biological activities, and development of novel therapeutic agents. Dr Raslan possesses extensive expertise in advanced spectroscopic techniques for the structural elucidation of natural compounds. She has actively participated in numerous research projects and scientific conferences.

Beyond Random Screening: MS/MS Molecular Networking and the Evolution of Natural Product Drug Discovery

Landmark advances in bioinformatics tools have recently broken through long-standing bottlenecks in untargeted mass spectrometry, transforming the landscape of natural products research. Traditionally, discovery workflows have been severely hindered by the frequent rediscovery of known compounds. While modern liquid chromatography-tandem mass spectrometry (LC-MS/MS) captures thousands of metabolic profiles simultaneously, translating these massive datasets into actionable structural insights represents a major technical challenge.

This presentation highlights MS/MS molecular networking as a transformative computational framework causing a true revolution in the field of natural product drug discovery. By employing vector-based spectral matching, molecular networking organizes and visualizes complex mass spectrometry data into distinct clusters, revealing underlying structural relationships based on fragmentation similarities. This powerful framework enables the rapid identification of known compounds within complex mixtures, significantly streamlining the dereplication process.

Crucially, a cluster anchored by a known compound identified via spectral library matching simultaneously highlights associated nodes representing previously uncharacterized structural analogues. This inherent capability to visually map the intricate chemical landscape of complex samples allows researchers to expedite the prioritization of their isolation workflows toward entirely new molecular families. Ultimately, this talk will discuss the foundational concepts of molecular networking, its underlying construction metrics, and its diverse pharmaceutical applications. Furthermore, we will explore the Global Natural Products Social Molecular Networking (GNPS) platform, illustrating how this widely utilized ecosystem is practically deployed to turn raw spectral data into targetable chemical insights.

Ass. Prof. Rehab Fikry Taher Fareed

National Research Centre, Egypt



Dr. Rehab Fikry is an associate professor of Pharmacognosy at the National Research Centre. She holds a Ph.D. in Pharmaceutical Sciences from Cairo University, specializing in natural products chemistry and pharmacognosy. Her research focuses on the isolation, identification, and biological evaluation of bioactive compounds from medicinal plants, with particular expertise in advanced spectroscopic techniques and chromatographic analysis. Dr. Fikry has extensive academic and research experience, having held positions at the National Research Centre and several universities, including Heliopolis University, Al-Ahram Canadian University and Egyptian Chinese University. She has authored numerous peer-reviewed publications in the fields of phytochemistry, pharmacology, and natural product-based therapeutics. Her work contributes to the discovery of plant-derived compounds with potential applications in treating diseases such as cancer, liver disorders, and metabolic conditions.

The Molecular Mechanisms Underlying the Antineoplastic Profile of Vitexin 2''-O- α -L-Rhamnopyranoside From *Cordyline australis* (G.Frost.): Isolation and Constituent Profiling

Cordyline australis (G.Forst.) Endl. Red Star leaves have been used in traditional medicine for several disorders. This study investigated the anticancer potential of *C. australis* leaves extract and characterized its bioactive constituents. Three compounds, a steroidal saponin, fruticoside K 23, and two flavonoids, vitexin 2''-O- α -L-rhamnopyranoside 35 and helichryoside 68, were isolated and identified from its aqueous methanolic extract (CAME) using column chromatography. Seventy-nine additional compounds were tentatively identified using high-performance liquid chromatography coupled with electrospray ionization mass spectrometry (HPLC-ESI-MS/MS) analysis and The Global Natural Product Social Molecular Networking (GNPS-MN), with 57 compounds reported for the first time in the *Cordyline* genus. CAME and its isolated compounds were evaluated for cytotoxicity by MTT assay against seven different cancer cell lines. Vitexin 2''-O- α -L-rhamnopyranoside 35 exhibited significant cytotoxicity against HCT-116 colon cancer cells. Additionally, CAME, fruticoside K 23, and vitexin 2''-O- α -L-rhamnopyranoside 35 demonstrated significant cytotoxicity against osteosarcoma (HOS) cells. Afterwards, the safety profile of CAME and all the isolated compounds were examined upon human normal cells BJ-1. At 100 μ g/mL, CAME and all isolated compounds showed a safe response on human normal BJ-1 cells (0.6%–8.5% cytotoxicity). Vitexin 2''-O- α -L-rhamnopyranoside 35 possessed the most significant selective anticancer response on osteosarcoma cells (HOS), with the least IC₅₀ value of 43.7 μ g/mL. It induced apoptosis in HOS cells by modulating Bax, Bcl-2, and caspase-3 expression and caused G1 phase cell-cycle arrest. These results highlight *C. australis* as a source of potential anticancer agents, particularly vitexin 2''-O- α -L-rhamnopyranoside 35, which warrants further investigation for its therapeutic potential.

Ass.Prof. Zeinab Abd Elkhalek Ali Elshahid

**Chemistry of Natural and Microbial Products, Pharmaceutical Drug Industry and
Research Institute, National Research Centre, Egypt**



Associate professor of Therapeutic Biochemistry at Chemistry of natural and microbial products department, Natural Research center, PhD in Biochemistry, faculty of science, Ain Shams University. I'm Interested in the search for new drug candidates from natural resources with variable biological activities as antitumor, antimicrobial, antioxidant, the treatment of diseases like: Al-zaheimer, Diabetes and Skin Hyperpigmentation. I participated in several national and international research projects funded by NRC and ASRT. I'm the principal investigator of a research project, entitled (An approach towards the isolation of novel bioactive secondary metabolites from some endophytic and symbiotic fungi as a new therapeutic agent for the treatment of type II diabetes and its complications), funded by national research center. Member of several reputable organizations, Arab Medical Association Organization, the National Society of Human Genetics (NSHG), Arab Society for Medical Research, Alzheimer Association International Society to Advance Alzheimer's Research and Treatment (ISTAART), Women In Science Without Borders (WISB), OWSD, American Society for Microbiology, and the Egyptian Chemical Society. ORCID ID: <https://orcid.org/0000-0001-8237-9568> . Publications: 38 original research articles in highly respected peer-reviewed international scientific journals. Scopus h-index = 10 and total citation of 250. Google Scholar h-index=11, total citation 323.

Investigation of Endophytic Fungi Secondary Metabolites with Potential Biological Activity

Overview: The quest for novel compounds of natural origin has been a persistent ambition for pharmaceutical research. Natural compounds are the main source of active ingredients in medicines, and in spite of modern pharmaceutical synthesis techniques, natural products are still the basis of almost half of all approved drugs. This success is in part due to the fact that natural products usually display high bioavailability and high affinity to target, as well as a minor loss of entropy when binding to proteins. Many endophytes have the potential to synthesize various bioactive metabolites that may directly or indirectly be used as therapeutic agents against numerous diseases. Moreover, marine biological systems still represent an endless reserve of vast natural bioactivities and massive chemical entities that yield a characteristic source of chemical species exhibiting potential biological activities. Numerous studies have discovered bioactive marine-associated fungi and their metabolites. Such bioactive products were evaluated for their antioxidant, radical scavenging, antimicrobial, and anticancer activities, among other biological activities. In this respect, the present work aimed to screen for bioactive secondary metabolites from fungal isolates isolated from some plants. These secondary metabolites will be assessed using various in vitro and in vivo systems, including antimicrobial, anti-inflammatory, and anti-cancer activities. The most active isolates that showed the highest bioactivity were selected for further studies. The potent isolates were subjected to large-scale fermentation on different media. Bioactivity measurements, chemical characterization, and structure elucidation will take place.

Associate Professor Dr. Gehan Fawzy Abdel Raoof Kandeel

**Associate Professor at Pharmacognosy Department, Pharmaceutical and Drug
Industries Research Institute, National Research Centre**

Dr. Gehan Fawzy Abdel in 2015 from Faculty of She is an Associate Professor Pharmaceutical and Drug National Research Centre focuses primarily on isolation, natural compounds from using advanced techniques for NMR, ¹H-¹H COSY, ¹³C-HSQC, HMQC), and bioactive natural products for use in has published many papers in international journals and has been serving as a reviewer and editor board member in many journals. She is a member of Organization for Women in Science for The Developing World (OWSD), a member of International Scientific Committee of World Academy of Science, Engineering and Technology (WASET).



Raoof, she completed her PhD Pharmacy, Cairo University. at Pharmacognosy Department, Industries Research Institute, Giza, Egypt. Her research purification and identification of medicinal plants, and marine identification (UV, IR, ¹H-NMR, DEPT ¹³C-NMR, assays in vivo and in vitro in treating different diseases. She

From Waste to Wellness: Bioactive Profiling and Multi-Target Therapeutic Potential of Vicia faba Peel By-Products

Plant by-products are increasingly recognized as sustainable sources of valuable bioactive compounds. The current study aims to investigate the bioactivities and metabolomic profile of the 70% methanol extract of Vicia faba peels. Comprehensive phytochemical profiling using ultra-high-performance liquid chromatography (UHPLC) hyphenated with quadrupole-time-of-flight tandem mass spectrometry (QTOF-MS) led to the tentative identification of 91 metabolites, most of which are reported in the tested plant for the first time. Furthermore, rhamnetin, quercitrin, rutin, and catechin were isolated from the tested extract and identified by different spectroscopic methods. Biological evaluations revealed an excellent safety

profile, as both the crude extract and the isolated compounds exhibited no in vitro cytotoxicity against normal skin fibroblast cells (BJ-1). The extract exhibited significant anticholinesterase activity, while rutin showed the highest antioxidant potential ($90.54 \pm 0.73\%$) in the DPPH assay, alongside potent inhibition of carbohydrate-metabolizing enzymes. These findings suggest that *V. faba* peels represent a promising, safe agricultural by-product with multi-target therapeutic potential, particularly for managing diabetes, oxidative stress, and neurological disorders.

Dr. Dina Magdy Ghanem

**Department of Pharmacognosy, Pharmaceutical and Drug Industries Research
Institute, National Research Centre, Egypt.**



Dr. Dina M. Ghanem is a pharmacognosy researcher at the Pharmaceutical and Drug Industries Research Institute, National Research Centre (NRC), Egypt. She earned her Ph.D. in Pharmaceutical Sciences (Pharmacognosy) from Cairo University in 2025, following her M.Sc. degree in 2016 and B.Sc. degree with honors in 2005. Her doctoral research focused on the pharmacognostical study of **Carissa carandas** and **Carissa macrocarpa** cultivated in Egypt. With nearly two decades of research experience, Dr. Ghanem specializes in medicinal plants, natural products chemistry, ethnopharmacology, and pharmacognostic evaluation. Her expertise encompasses phytochemical screening, metabolite profiling, extraction techniques, chromatographic analysis, and the isolation and structural elucidation of bioactive compounds using advanced spectroscopic and analytical methods. She is also actively involved in biological activity assessment, metabolomics, molecular docking, and computer-aided drug discovery. Dr. Ghanem has contributed to several peer-reviewed scientific publications addressing the phytochemistry and pharmacological potential of medicinal plants. Her research has focused on identifying natural bioactive compounds with promising therapeutic applications and exploring sustainable plant-based resources for drug discovery and development. She is a member of the Special Unit of Pharmacognosy and Chemistry of Medicinal Plants, the National Laboratory for Algae, and the Incubator for Nutritional Supplements and Pharmaceutical Preparations from Medicinal Plants at the National Research Centre. In addition, she actively participates in scientific conferences, workshops, and educational programs, contributing to the training of university and school students. Her current research interests include medicinal plants, metabolomics, natural product chemistry, molecular networking, pharmacological evaluation of plant-derived compounds, and computational approaches for modern drug discovery.

Integrated Phytochemical Profiling and Biological Evaluation of Egyptian *Tribulus terrestris* L.: A Promising Source for Therapeutic Product Development

The genus *Tribulus* of the Zygophyllaceae family includes numerous species that grow as shrubs or herbs in subtropical regions around the world. Aerial parts of *Tribulus terrestris* L. growing in Egypt were successively extracted with solvents of increasing polarity. These

successive extracts were studied phytochemically and biologically to discover the importance of this plant as well as the relation between its secondary metabolites and the proven biological activities. Isolation and identification of the pure compounds from the biologically active fractions were carried out using different chromatographic and spectroscopic techniques. The biological activities of polar, non-polar, and saponin fractions were evaluated. GC/MS analysis of the volatile constituents revealed that methyl linolenate (18.56%) is the major compound. GLC analysis of the lipoidal matter revealed that heptadecanoic acid (33.56%) is the main component. HPLC analysis of carbohydrate and amino acid contents revealed the presence of inulin (5.61%) and glutamic acid (2.85%) as the major components, respectively. Phytochemical investigation of the biologically active aqueous methanolic extract revealed the presence of phenolic acids (11.16%), flavonoids (6.076%), as well as steroidal saponins (7.38%). Purification of the aqueous methanolic extract led to isolation and identification of rutin, quercetin and diosgenin. The results of the biological investigation of both polar and non-polar fractions revealed their aphrodisiac, antimicrobial, cytotoxic and antioxidant activities with remarkable effects. The herb in this study showed potent and noteworthy biological activities due to the presence of various active secondary metabolites.

Ass. Prof. Marwa N. Ahmed

School of Biotechnology, Nile University, Egypt



Dr. Marwa Nabil Ahmed is currently an assistant professor of Microbiology at the Biotechnology School, Nile University. Her broad expertise includes microbial biotechnology, antimicrobial resistance, bacterial biofilms and industrial microbiology. She earned her master's degree in biotechnology from Georgia State University, USA, where she also worked as a teaching and research Assistant, contributing to undergraduate education and research training. She earned her PhD degree in 2019 in Immunology and Infectious Diseases from the Faculty of International Health and Immunology, University of Copenhagen, Denmark, where she studied the molecular mechanisms of biofilm resistance in *Pseudomonas aeruginosa* at the phenotypic, genomic, transcriptomic levels. Her research on the intersection of environmental and genetic factors on bacterial adaptation and pathogenesis under antibiotic stress has led to published works in high-impact journals. Her translational achievements include a U.S. patent, "Bactericidal Compositions and Methods for Treating Pathogenic Biofilms". In industrial biotechnology, Dr. Marwa has optimized microbial production of value added bioproducts from agro-industrial wastes. Additionally, she has implemented her bioinformatics expertise to elucidate the evolutionary trajectories of antibiotic resistance in pathogenic bacteria and to identify novel genetic pathways associated with the biosynthesis of valuable microbial bioproducts. As Co-Principal Investigator of the STDF One Health Research and Innovation Grant "Bacteriophages as a Potential Alternative Approach to Combating Extended Spectrum Beta-Lactamases Producing Bacteria," she continues

advancing research on epidemiological surveillance, phage therapy and bioinformatic analysis. She has been supported by prestigious funding agencies, including the Georgia Research Alliance Venture Projects.

Evolutionary Trajectories of Antibiotic Resistance in *Pseudomonas aeruginosa*: Mechanisms, Dynamics, and Strategies for Constraint

Chronic bacterial infections, exacerbated by antibiotic resistance, pose a significant health challenge, particularly in the context of *Pseudomonas aeruginosa* (Pa) infections in Cystic Fibrosis (CF). The formation of biofilms by Pa complicates treatment, shielding bacteria and contributing to antibiotic resistance. Ciprofloxacin is a commonly used antibiotic for Pa in CF patients. Experimental evolution studies of PAO1 wild-type (WT) and Δ katA in sub-inhibitory ciprofloxacin conditions revealed that biofilm growth promotes the development of low-level resistant mutants. Oxidative stress was found to increase mutagenesis and resistance development. Consequently, we explored the genome and transcriptome of the evolved PAO1 and Δ katA colony biofilm and planktonic culture to elucidate the phenotypic, genomic and transcriptomic changes leading to antibiotic resistance evolution. Additionally, we investigated different strategies and formulations to constrain the evolution of antibiotic resistance including cellular hysteresis, antioxidants and postbiotics. The transcriptomic analysis of PAO1 and Δ katA evolved populations indicates higher gene expression in CIP-evolved populations than CTRL populations. Similarly, biofilms (both CIP and CTRL) exhibit more expressed genes than planktonic populations, with minimal overlap between biofilm and planktonic conditions. Furthermore, cellular hysteresis strategy, application of postbiotics and antioxidants showed a significant reduction in the evolution rate of antibiotic resistance. Overall, this comprehensive analysis enhances our understanding of the intricate genomic and transcriptomic changes in Pa populations under ciprofloxacin exposure, shedding light on both shared and population-specific adaptive responses.

Ass. Prof. Rasha Mohamed Hassan

at Medicinal and Pharmaceutical Chemistry Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Cairo, Egypt



Dr. Rasha is currently an Associate Professor of Pharmaceutical Chemistry at Medicinal and Pharmaceutical Chemistry Department, Pharmaceutical and Drug Industries Research Institute, National Research Centre, Cairo, Egypt. She obtained M.Sc. from Faculty of Pharmacy, Ain Shams University (2011) and Ph.D. from Faculty of Pharmacy, Cairo University (2015). Her publications cover a variety of drug-related areas in central nervous system, cancer, bacterial resistance, inflammatory and metabolism disorders in international peer-reviewed journals. She is in-charge of several valuable pharmaceutical research projects funded by National Research Centre and Science, Technology and Innovation Funding Authority (STDF), Egypt.

Synthetic Anti-Biofilm Agents as a Strategy in Combating Bacterial Resistance

The great increase in the bacterial resistance to the known antibiotics is considered a major obstacle in the control of many infections. Bacterial biofilm formation is regarded as a key element in bacterial resistance and has a substantial impact on the bacteria's ability to survive in host cells. Antibiotics affect bacterial cells in biofilms very differently than they do free-floating planktonic cells of the same strain. The anti-biofilm agents that could inhibit the biofilm production without affecting the bacterial growth, apply less selective pressure over the bacterial strains than the traditional antibiotics; thus the development of bacterial resistance would be less frequent. In this study, new quinazoline derivatives were synthesized and screened for their anti-bacterial effects. Some of the synthesized compounds inhibited biofilm formation in *Pseudomonas aeruginosa*, which is regulated by quorum sensing system, at sub-minimum inhibitory concentrations. These compounds decreased the exopolysaccharide production which is the major component of the matrix binding biofilm components together. Also, at sub-MICs *Pseudomonas* cells twitching motility was impeded by these compounds, a trait which augments the cells pathogenicity and invasion potential. Molecular docking study was performed to further evaluate the binding mode of the compounds as inhibitors of *P. aeruginosa* quorum sensing transcriptional regulator PqsR. The achieved results demonstrate that achieved compounds bear promising potential for discovering new anti biofilm and quorum quenching agents against *Pseudomonas aeruginosa* without triggering resistance mechanisms as the normal bacterial life cycle is not disturbed.

Ass. Prof. Doaa Khalaf

Microbiology and immunology department, Veterinary Research Institute, National Research Centre (NRC), Egypt



Dr. Doaa Diab Khalaf is an Associate Professor of Microbiology and Immunology at the National Research Centre (NRC) in Egypt. With a Ph.D. in Veterinary Medicine from Sadat City University, her professional focus lies in applied microbiology, food safety, and the development of diagnostic tools for veterinary and food industry applications. Throughout her career, Dr. Khalaf has contributed to various national research initiatives, participating in projects that explore the characterization of bacteriocins for food biopreservation, the development of diagnostic kits for meat adulteration, and the application of nanotechnology in managing cattle mastitis. Her research interests also extend to the study of antimicrobial resistance and the identification of pathogenic microbes in animal and food products, with a focus on translating laboratory findings into practical solutions for agricultural sustainability. In addition to her research, Dr. Khalaf is engaged in the institutional framework of the National Research Centre, participating in committees related to technology incubation and entrepreneurship. She is committed to the application of modern technical tools, such as AI, within the animal sector and actively participates in knowledge-sharing through scientific workshops and training sessions for graduate students. As a researcher, she is dedicated to the ongoing advancement of scientific methodology and the role of women in the scientific community. Her work aims to contribute to the development of animal wealth and public health through rigorous scientific inquiry and the exploration of new avenues in microbiology and innovation.

Antimicrobial Resistance in the Animal Sector: The 'One Health' Approach as a Sustainable Framework

Antimicrobial resistance (AMR) represents a growing global threat to food security and public health, necessitating the adoption of a "One Health" framework to integrate efforts across animal, human, and environmental health. The presentation addresses the scientific dimensions of microbial resistance in the animal sector, highlighting the importance of transitioning from total reliance on conventional antibiotics toward developing sustainable alternatives. The presentation examines innovative microbiological tools, such as the use of natural extracts and nanotechnology applications, as therapeutic alternatives that contribute to reducing the selection pressure leading to resistant strains. Furthermore, the abstract discusses the strategic role of rapid molecular diagnostics in enhancing herd management efficiency, thereby minimizing indiscriminate drug use and promoting sustainable animal production. Additionally, the presentation emphasizes that addressing this phenomenon requires an integrated scientific ecosystem that combines innovation in biological sciences with technological entrepreneurship. By fostering interdisciplinary collaboration and utilizing modern technologies, this talk aims to provide a practical roadmap to support the Sustainable Development Goals and ensure the preservation of biological resources for future generations.

Ass.Prof. Islam Hany Ali Haridy

National Research Centre (NRC) - Cairo – Egypt



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New microtubule modulators: Design, Synthesis and Anticancer Evaluation

New chalcone, and 4,5-dihydropyrazole compounds were designed and synthesized. A molecular docking study was done for the synthesized compounds at tubulin binding site. The newly synthesized compounds were tested for their in vitro anticancer screening to determine their inhibitory activities against a panel of cancer cell lines at 10 μ M concentration at National Cancer Institute (NCI) - US. Tubulin polymerization assay was investigated for the synthesized compounds at 50 μ M and compared with CA-4 and paclitaxel as standards. All the compounds showed microtubules targeting activity. Then, a tubulin immunohistochemical study was performed to the most active compounds to confirm their effects on tubulin polymerization via immunofluorescent detection. Cell cycle analysis and apoptosis were done via Flow cytometry for the tested compounds. The acute toxicity was performed to the most active compounds, in mice and then subjected to anticancer evaluation testing in rats. The anticancer parameters were determined in addition to histopathological examination. In conclusion, 4,5-dihydropyrazoles presented in vitro and in vivo anticancer activities without histopathological alterations.

Keywords:

Chalcones, dihydropyrazoles, microtubules, modulators, tubulin polymerization.

Session 4 Medical sciences and Biology

Prof. Eman E. El Shanawany

Parasitology and animal diseases department, Veterinary Research Institute, National Research Centre (NRC), Egypt



Prof. Eman E. El Shanawany is Professor of Immunology and Parasitology at the Veterinary Research Institute, National Research Centre (NRC), Egypt, where she also serves as Technical Manager of the ISO/IEC 17025 Accredited Immunology and Parasitology Laboratory. She was a Visiting Research Fellow at the National Institute of Parasitic Diseases, Chinese Center for Disease Control and Prevention (China CDC), Shanghai, China, and completed an Erasmus International Credit Mobility teaching and research visit at Dublin City University, Ireland, where she taught undergraduate courses in cell culture and immunology. Her research focuses on host–parasite interactions, immunology, vaccine development, immunodiagnostics, cancer immunotherapy, tropical diseases, and One Health. She has served as Principal Investigator and Co-Investigator in several nationally and internationally funded projects, including vaccine development against toxoplasmosis, and international collaborative programs with China CDC. She also contributed to the development of a commercial glycoprotein-based diagnostic kit for fasciolosis. Prof. El Shanawany is the recipient of the National Research Centre Prize in Immunology Research (2023) and the National Research Centre Best PhD Thesis Award in Veterinary Sciences (2013). She has published extensively in international peer-reviewed journals, serves as a reviewer for several SCI-indexed journals, and is actively involved in scientific conference organization, postgraduate supervision, and laboratory quality management. She is a member of several professional scientific societies and is committed to advancing translational research in parasitology and immunology.

Platyhelminth Glycans: From Host–Parasite Interactions to Diagnosis, Immunomodulation, and Vaccinology

Platyhelminths (Phylum Platyhelminthes, Subphylum Neodermata) comprise a diverse group of parasites of considerable veterinary, medical, and economic importance. Their glycans are expressed as components of O- and N-linked glycoproteins, glycolipids, and excretory/secretory products, as well as on the parasite surface, where they play essential roles in host–parasite interactions. These glycoconjugates interact with antigen-presenting cells through specific lectin receptors, particularly on dendritic cells, leading to modulation of the host immune response toward a T helper 2 (Th2)-dominant profile that promotes parasite survival. The adaptive immune response against platyhelminth glycans is characterized by the

production of specific antibodies, including IgG and IgE, together with Th2-associated cytokines such as IL-4, IL-5, and IL-13. Anti-glycan antibodies represent valuable biomarkers and are widely exploited for the serodiagnosis of platyhelminth infections. Moreover, parasite glycoproteins and glycolipids have emerged as promising immunodiagnostic targets for the detection of infections in both humans and livestock, with their diagnostic performance largely depending on the structural characterization of glycan epitopes. Beyond diagnosis, the immunomodulatory properties of platyhelminth glycans have highlighted their potential in vaccine development and as novel therapeutic candidates for autoimmune and inflammatory diseases. This presentation discusses the biological significance of platyhelminth glycans and reviews their current and potential applications in diagnosis, immunomodulation, and vaccinology.

Dr. Aya abdallah seleem

Faculty of Veterinary Medicine Cairo University, Egypt



Dr. Aya Abdallah Seleem is a Lecturer of Zoonoses at the Faculty of Veterinary Medicine, Cairo University, Egypt. My expertise lies in molecular biology and antimicrobial resistance, with a research focus on zoonotic pathogens at the human–animal interface. My work explores the molecular epidemiology, detection, and characterization of infectious agents of veterinary and public health importance. My research interests include antimicrobial resistance profiling, molecular diagnostics of zoonotic bacteria, and surveillance of emerging and re-emerging zoonotic diseases within a One Health framework. She is particularly interested in understanding transmission dynamics between animals and humans and advancing evidence-based strategies for infection control and antimicrobial stewardship in veterinary settings. In addition to research, I teach zoonotic diseases and veterinary public health and is actively involved in training veterinary students in molecular and epidemiological approaches to infectious disease investigation. She is committed to promoting interdisciplinary collaboration and strengthening research capacity in zoonoses and antimicrobial resistance in the region. I am also engaged in science communication and professional coaching initiatives that support awareness, resilience, and personal development among women in scientific and academic environments.

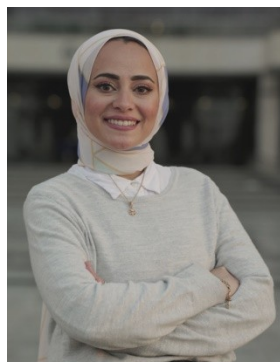
The Rise of Pan-Drug Resistance: Implications for Public Health and zoonotic diseases

Antimicrobial resistance (AMR) poses a global health threat. Pan-drug resistant (PDR) microbes, being resistant to all the available antimicrobial agents, are the most objectionable. Overuse and misuse of antibiotics in clinical, agricultural, and animal health environments have encouraged the emergence and spread of multidrug-resistant (MDR), extensively drug-resistant (XDR), and pandrug-resistant (PDR) strains of bacteria. Genetic mutations, plasmid-mediated gene transfer, enzymatic degradation of antibiotics, efflux pump overexpression,

reduced membrane permeability, and modifications in target pathways all play a role in such mechanisms of resistance. Zoonotic bacteria only make it worse when strains of resistance spread further within humans as well as animals. The alternative options of classical therapy are declining, while new antibiotics are emerging at a slow pace. Hence, other methods are gaining prominence ever more. Examples include the utilization of artificial intelligence (AI) for the identification of drugs, use of antimicrobial peptides (AMPs) as the natural defense agents, and isolation of phytochemicals from medicinal plants possessing antibacterial and resistance-modulating activity. This work discusses the molecular mechanisms of resistance, global burden of antimicrobial resistance (AMR), and new interdisciplinary approaches to tackle pandrug-resistant (PDR) bacteria. To address this multifaceted issue, a need exists to enhance antimicrobial stewardship, augment surveillance efforts, and promote research on bioactive natural products. Global action is called for to maintain the efficacy of antimicrobial therapy, encompassing scientific discovery, regulatory changes, and public engagement in health initiatives.

Dr. Rana Ahmed Mohamed Eissa

Badr University, Egypt



Rana Ahmed Eissa is a Research Assistant at the Research Center, Badr University in Cairo (BUC). She earned her Bachelor's degree in Pharmacy from Ain Shams University in 2018. In 2020, she got her M.Sc. degree in Biotechnology from the School of Science and Technology, Nottingham Trent University, Nottingham, UK. She is currently a Ph.D. candidate in the Department of Microbiology and Immunology, Faculty of Pharmacy, Ain Shams University. In 2023, she was awarded the L'Oréal–UNESCO For Women in Science Egypt Young Talents Fellowship in the doctoral category. Her research interests focus on the formulation and evaluation of controlled-release drug delivery systems and microbiology, molecular biology, and biotechnology techniques. She contributed as co-author of several publications in peer-reviewed journals. She has also participated in research projects funded by the Egyptian Academy of Scientific Research and Technology and the Science, Technology and Innovation Funding Authority. She has been actively involved in delivering annual summer training programs for undergraduate students. She has also supervised two undergraduate graduation projects at the School of Biotechnology for the Classes of 2024 and 2025. In October–November 2024, she completed a one-month research visit to the Microbiology Laboratories at the School of Science and Technology, Nottingham Trent University. During this period, she gained hands-on experience in molecular biology techniques, including RNA extraction, cDNA synthesis, and real-time PCR to analyze. This experience further enhanced her skills in experimental optimization and gene expression data analysis.

Antibacterial Film-Coated Stainless Steel Trays for Controlling *Klebsiella pneumoniae* and *Pseudomonas aeruginosa* in Intensive Care Units

Infections caused by multidrug-resistant (MDR) bacteria in intensive care units (ICUs) are a leading cause of global morbidity and mortality, placing a significant burden on healthcare systems managing critically ill patients. Gram-negative pathogens, including *Klebsiella pneumoniae* and *Pseudomonas aeruginosa*, are highly prevalent on stainless steel trays in ICUs, and their ability to adhere to medical devices poses a serious threat to human health. Moreover, these pathogens often exhibit resistance to conventional antimicrobial therapies. Nanotechnology, combined with antibiotic therapy, has emerged as a promising alternative of conventional methods to manage MDR infections.

In this study, a novel system was developed to control the growth of *K. pneumoniae* and *P. aeruginosa* on stainless steel trays in ICUs. An antibacterial chitosan-silver nanoparticle (AgNP) film was synthesized and further loaded with ciprofloxacin and vancomycin. The nanocomposite film exhibited a mean number-averaged hydrodynamic diameter of 202 ± 3.46 nm, which was confirmed by scanning electron microscopy, showing well-defined nanovesicles evenly distributed throughout the film. In vitro characterization of the antibacterial film was then performed, including evaluation of swelling ratio, moisture content, and Fourier transform-infrared spectroscopy. The antimicrobial activity of the nanocomposite film loaded with vancomycin and ciprofloxacin demonstrated significant inhibition of *K. pneumoniae* and *P. aeruginosa* compared to the individual antibiotics alone. The successful application of this antibacterial chitosan-AgNP film could add new infection control strategies, improve patient outcomes and reduce the burden of healthcare-associated infections.

Reham Amgad Mohamed-Ezzat

Associate Professor of Therapeutic

**Chemistry at Chemistry of Natural & Microbial Products Department,
Pharmaceutical & Drug Industries Research Institute, National Research Centre,
Cairo, Egypt.**



Reham Amgad Mohamed-Ezzat, PhD in Organic Chemistry, Associate Professor of Therapeutic Chemistry at Chemistry of Natural & Microbial Products Department,

Pharmaceutical & Drug Industries Research Institute, National Research Centre, Cairo, Egypt. She got her BSc (2010) in Chemistry & Biochemistry-Faculty of Science. She was awarded her MSc (2015) & PhD degree of science in organic chemistry in 2019 from Helwan University. She has published books and articles in international journals in the field of bio-organic and medicinal chemistry. She also published many conferences presentations. She has experience in the field of design and synthesis of new molecules of medicinal application, as well as the implementation of novel strategies in the syntheses of bioactive compounds. She is a member in the Egyptian Chemical Society, and in the Egyptian Society for Basic Sciences. E-mail: reham_amgad_2010@yahoo.com; ra.mohamed-ezzat@nrc.sci.eg. Phone: 01220613360 Scientific Achievements:

H-INDEX 14 No. of publications: 38 <https://www.scopus.com/authid/detail.uri?authorId=55502779900> Last published book in Elsevier. New Strategies Targeting Cancer Metabolism Anticancer Drugs, Synthetic Analogues and Antitumor Agents 1st Edition - June 7, 2022 Latest edition Imprint: Elsevier Authors: Galal H. Elgemeie, Reham A. Mohamed-Ezzat Language: English Paperback ISBN: 9780128217832 - eBook ISBN: 9780128217849

Synthetic strategies towards Novel sulfonamides as Potentially Active Agents

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New synthetic approaches were developed for the synthesis of novel sulfonamides as potent therapeutic agents, utilizing various sulfaguanidine derivatives. Additionally, a new class of pyrimidine sulfonamide thioglycosides was successfully designed and synthesized. The newly obtained substituted pyrimidine sulfonamides, reported here for the first time, were produced in high yields and demonstrated promising broad-spectrum biological activities, including antitumor, antimicrobial, and antiviral effects. The in vitro antiproliferative activity of most synthesized compounds was evaluated using the NCI-60 human cancer cell line panel. Furthermore, the antimicrobial potential of these compounds was also investigated. The antiviral activity of the compounds against SARS CoV-2 virus was investigated utilizing MTT cytotoxicity assay to estimate the half-maximal cytotoxic concentration (CC₅₀) and inhibitory concentration 50 (IC₅₀). The most active compound revealed potent anti-viral potency against SARS-CoV-2 comparable to that of remdesivir. Our results demonstrate that these novel pyrimidine sulfonamides could potentially serve as novel anti-viral drugs.

Keywords:

Synthesis, pyrimidines, Sulfonamides, SARS-CoV-2.

Ass.Prof. Iman Ahmed Youssef Ghannam

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Dr. Iman Ahmed Youssef Ghannam is an Associate Professor of Pharmaceutical Chemistry - Chemistry of Natural and Microbial Products Department - Pharmaceutical and Drug Industries Research Institute - National Research Centre (NRC) - Cairo - Egypt since January 2023. Dr. Ghannam has her area of expertise in the development and rational design of novel motifs targeting cancer and metabolic and neurodegenerative disorders including hyperlipidemia, Type II diabetes mellitus, and Alzheimer's disease (AD). Dr. Ghannam has joined as a principal investigator (PI), Co-PI, and member in several research projects funded from National Research Centre (NRC) and Science and Technology Development Fund (STDF) - Egypt since 2015. Dr. Ghannam has also participated as a PI for a joint collaboration with Consiglio Nazionale delle Ricerche (CNR) - Italy through a joint NRC-CNR bilateral co-funded research project with Dr. Francesca Sciandra in 2019. Dr. Ghannam has co-supervised M.Sc. and PhD students from Faculty of Pharmacy - Cairo University since 2015 till present. Dr. Ghannam co-authored several publications including research papers and review articles indexed in (SCOPUS) in highly impacted journals in medicinal chemistry since 2012, and also peer-reviewed for several manuscripts in several Journals; Journal of Medicinal Chemistry, European Journal of Medicinal Chemistry, Bioorganic Chemistry, European Journal of Medicinal Chemistry Reports, Archiv der Pharmazie, Scientific Reports, and ACS Chemical Neuroscience, and Molecular Diversity.

Development of Benzenesulfonamide-based PPAR γ Agonists for the Treatment of Type II Diabetes Mellitus

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Abstract:

Peroxisome proliferator-activated receptors (PPAR) isoforms (α , γ , and β/δ) play a crucial regulatory role in glucose, lipid metabolism. Thiazolidinone-benzenesulfonamide and chalcone-benzenesulfonamide hybrids were designed, synthesized and tested *in vitro* for their PPAR γ agonistic activity compared to pioglitazone via determination of their EC₅₀ values. Immunohistochemical studies were performed in HepG-2 cells to confirm the PPAR γ protein expression for the most active compounds. Moreover, the most active compounds were tested for their selectivity on PPAR α/δ isoforms. *In vivo* studies revealed that the most compounds possessed antidiabetic activity better than that of pioglitazone. Furthermore, molecular docking studies and molecular dynamic simulations were carried out for the newly synthesized compounds to study their predicted binding modes and energies in the PPAR γ binding site.

Ass. Prof. Amira Samy Talaat Abou Taleb

Electronics Research Institute, Cairo, Egypt



Amira Samy Talaat is an Assistant Research Professor at the Computers and Systems Department, Electronics Research Institute, Cairo, Egypt. Her research interests include artificial intelligence, machine learning, explainable artificial intelligence, computer vision, intelligent systems, and their applications in healthcare and engineering. Her work focuses on developing transparent, reliable, and human-centered artificial intelligence methods that support informed decision-making and improve understanding of complex machine-learning models. She is particularly interested in explainable artificial intelligence techniques that help researchers and professionals understand why an AI system produces a specific prediction. Her research also addresses model reliability, data quality, fairness, and responsible artificial intelligence. She has participated in multidisciplinary research projects and scientific publications involving medical data analysis, predictive modeling, and interpretable artificial intelligence. Through her academic and research activities, she aims to promote the responsible use of artificial intelligence and strengthen collaboration between computer scientists, engineers, healthcare professionals, and researchers from different disciplines. Her participation in the World Forum for Women in Science reflects her interest in international scientific exchange, knowledge sharing, and the contribution of women researchers to sustainable development and technological innovation.

Beyond the Black Box: How Explainable AI Can Support Trustworthy Healthcare Decisions

Artificial intelligence is increasingly used in healthcare to support disease diagnosis, risk prediction, medical imaging, treatment planning, and patient monitoring. However, many advanced artificial intelligence models operate as “black boxes,” providing predictions without clearly explaining how those predictions were produced. This lack of transparency may reduce the confidence of healthcare professionals and patients, particularly when artificial intelligence is used in high-stakes medical decisions. This presentation introduces Explainable Artificial Intelligence (XAI) as an approach for making artificial intelligence predictions more understandable, transparent, and useful to human decision-makers. It explains the difference between global explanations, which describe the overall behavior of a model, and local explanations, which clarify an individual prediction for a specific patient. The presentation also introduces commonly used explanation techniques, including SHAP and LIME, and demonstrates how they can identify the clinical variables that influence an artificial intelligence prediction. Practical applications of XAI in radiology, pathology, genomics, clinical decision-support systems, intensive care monitoring, mental health, and remote patient monitoring will be discussed. The presentation also addresses important limitations, including misleading explanations, bias, the trade-off between predictive performance and interpretability, and the need for explanations that are meaningful to healthcare professionals rather than merely technically accurate. Explainable AI should not be considered a replacement for medical expertise. Instead, it can serve as a communication and decision-support tool that enables clinicians to examine, question, and appropriately use artificial intelligence predictions. By promoting transparency, accountability, and informed human oversight, XAI can contribute to more trustworthy and responsible healthcare innovation.

Yosra Morsy

OASIS Psychology Creations, Founder&CEO Mental Healthcare in 2035:From Clinics to intelligent Care



Yosra M. Morsy is a licensed Clinical Psychologist and Psychotherapist with the Egyptian Ministry of Health and a Ph.D. candidate in Clinical Psychology. She holds a Master's degree in Clinical Psychology and has extensive experience providing psychological assessment and psychotherapy for children, adolescents, adults, and older adults using evidence-based approaches. Yosra is the Founder and CEO of Oasis Psychology Creations, an initiative dedicated to advancing mental health through research, innovation, education, and interdisciplinary collaboration. Her professional interests focus on digital mental health, artificial intelligence in psychological practice, behavioral science, ethics in emerging technologies, and the future of mental healthcare. She is actively involved in international scientific collaborations and has presented her work at conferences focused on psychology, digital health, and innovation. Her current research explores how technology can enhance psychological assessment and improve access to evidence-based mental healthcare while

maintaining ethical standards and human-centered practice. In addition to her clinical and research activities, Yosra is an international trainer and speaker who delivers workshops on psychology, mental health, and innovation. She is passionate about translating scientific knowledge into practical solutions that improve well-being and contribute to sustainable healthcare systems. Through her work, Yosra aims to bridge psychology with emerging technologies, fostering collaborations between clinicians, researchers, engineers, and entrepreneurs to shape the future of mental healthcare. Her long-term vision is to promote accessible, ethical, and innovative mental health services that create meaningful impact for individuals and communities worldwide.

Mental Healthcare in 2035: From Clinics to Intelligent Care

Mental health is becoming one of the world's most important public health priorities. However, the growing demand for mental health services is increasing faster than the availability of mental health professionals. This gap has encouraged researchers and innovators to explore new ways of delivering psychological care through technology while maintaining high standards of quality, ethics, and human connection. This presentation explores how mental healthcare may evolve by 2035, based on current scientific and technological trends. It discusses the growing role of artificial intelligence, digital mental health, virtual reality, wearable devices, digital therapeutics, and telepsychology in supporting prevention, early detection, assessment, treatment, and long-term follow-up. These innovations have the potential to make mental health services more accessible, personalized, and efficient for diverse populations. At the same time, the presentation emphasizes that technology should support—not replace—mental health professionals. Human empathy, ethical practice, clinical judgment, and the therapeutic relationship will remain the foundation of effective psychological care. The future of mental healthcare depends on successfully integrating technological innovation with evidence-based clinical practice. By bringing together psychology, healthcare, and innovation, this presentation provides a forward-looking perspective on how intelligent technologies can improve mental healthcare while preserving its human-centered values. The discussion contributes to Sustainable Development Goal 3 (Good Health and Well-Being) by highlighting innovative approaches to improving mental health services and expanding access to quality care.

Afnan Hamdy Elgowily
Postdoctoral Researcher , Chemistry Department, Biochemistry Section , Faculty of
science , Tanta University



Afnan Elgowily, PhD, is a postdoctoral researcher at the Autophagy Research Center, Department of Physiology, Juntendo University Graduate School of Medicine, Tokyo, and a Lecturer of Biochemistry at Tanta University, Egypt. She earned her PhD in Biochemistry through a joint program involving Niigata University, Juntendo University, and Tanta University, following earlier MSc and BSc training in biochemistry and chemistry. Her research focuses on autophagy, selective cargo recognition, protein quality control, cancer biology, and disease-associated cellular stress. She has extensive experience in molecular and cell biology, including CRISPR Cas9 gene editing, molecular cloning, protein purification, stable and transient cell engineering, mammalian cell culture, real-time PCR, immunoblotting, flow cytometry, ELISA, and animal experimentation. Her current work investigates how NBR1 and p62/SQSTM1 cooperate in selective autophagy and how their interaction regulates the organization, composition, and maturation of p62 bodies. Afnan has contributed to research published in Nature Communications, the Journal of Biological Chemistry, Cell Biochemistry and Function, Biomedicines, Nanomaterials, and other international journals. Her publications include studies on UFM1-dependent ER-phagy, phase-separated p62 condensates, cancer-associated autophagy, apoptosis, drug delivery, and therapeutic repurposing. She received a Chugai Foundation postdoctoral fellowship and an international Egyptian mission scholarship for research training in Japan. Alongside research, she has taught practical biochemistry and chemistry courses to undergraduate students since 2010. Her long-term goal is to advance mechanistic discoveries in autophagy and proteostasis toward improved understanding and treatment of human disease through international collaboration. She also values scientific mentoring, interdisciplinary teamwork, and building research partnerships between Japan, Egypt, and global institutions.

Pathophysiological Significance of Mallory Denk Bodies: Addressing a Century-Old
Mystery

Afnan Elgowily and Masaaki Komatsu

Department of Physiology, Juntendo University Graduate School of Medicine, Tokyo, Japan

Mallory Denk bodies (MDBs) are cytoplasmic inclusions frequently found in hepatocytes in chronic liver diseases, including alcoholic hepatitis, metabolic dysfunction-associated steatotic liver disease, cholestatic liver disease, and hepatocellular carcinoma. They contain

ubiquitin, p62/SQSTM1, and cytokeratin filaments and are thought to represent solidified p62 bodies membraneless condensates formed by p62 and polyubiquitinated proteins through liquid liquid phase separation that evade autophagic clearance. Beyond sequestering aberrant proteins for degradation, p62 bodies also regulate stress-responsive signaling, including NRF2 activation through KEAP1 sequestration. We previously showed that TBK1 phosphorylates p62 at Ser403, promoting condensate gelation and autophagic degradation. Here, we identify casein kinase 2 (CK2) as a previously unrecognized p62 kinase. AlphaFold3 modeling predicted a stable CK2 p62 interaction interface, and in vitro pull-down assays confirmed direct binding. Reconstitution assays demonstrated efficient recruitment of CK2 into p62 ubiquitin condensates. This recruitment was abolished by p62 interface mutations (L307A/M311A/I314A), despite preserved condensate formation. In vitro kinase assays showed that CK2A1 and CK2A2 phosphorylate p62 specifically at Ser349, but not Ser403; notably, only CK2A1 phosphorylated Ser349 within condensates. Because Ser349 phosphorylation enhances KEAP1 binding, CK2 recruitment is expected to promote KEAP1 retention and NRF2 stabilization. Immunofluorescence further confirmed CK2 localization to p62 bodies in cells. These findings define a CK2 p62 NRF2 signaling axis distinct from TBK1Ser403-mediated regulation of condensate material properties. CK2-dependent Ser349 phosphorylation may contribute to hepatocyte adaptation, MDB persistence, and chronic liver pathology, providing a framework for in vivo studies and therapeutic evaluation of CK2 p62 signaling in MDB-associated liver disease and clarifying the century-old pathophysiological significance of MDBs in humans.

Session 5 Medical sciences and Biology:

Prof. Rasha Mohamed Abdelraouf

Faculty of Dentistry, Cairo University, Egypt



Rasha Mohamed Abdelraouf is Professor of Biomaterials and Postgraduate Course Director at the Faculty of Dentistry, Cairo University. Her research focuses on dental biomaterials and innovative dental technologies. She is the recipient of the State Encouragement Award for Women in Health and Pharmaceutical Sciences (2024; awarded in 2025). She holds an Egyptian patent and an international Patent Cooperation Treaty (PCT) application and has published extensively in international peer-reviewed journals.

From Static to Smart: 4D Printing in Dentistry

The 4D printing, being an additive manufacturing technique, reduces waste and allows the fabrication of complex structures compared to subtractive manufacturing. The 4D printing is an advancement of already-existing 3D printing by including time. The 4D printing involves the use of 3D printing techniques for printing smart materials. The 4D structures can respond with time to stimuli to which they are exposed. They can respond to heat, water, light, electric energy, magnetic energy, stress, strain, pressure, etc. Several materials could be printed with this technology. The 4D printing has several applications, introducing smart functional biomaterial.

Dr. Tamer Mohamed Hamdy

**Associate Researcher Professor in the Department of Restorative and Dental Materials
Department, Oral and Dental Research Institute, National Research Centre (NRC)**



Dr. Tamer Mohamed Hamdy Mahmoud Ibrahim is an Associate Researcher Professor in the Department of Restorative and Dental Materials Department, Oral and Dental Research Institute, National Research Centre (NRC). He obtained his PhD in in Dental Materials Science from the Faculty of Dentistry, Cairo University. In 2019, he obtained certification as an international trainer from Ain Shams University, Cairo, Egypt. Serves as an international peer reviewer for several international dental journals. **Dr. Tamer M. Hamdy** received the National Research Centre Award for Best International Scientific Publication for Assistant Researchers in 2015, followed by the NRC Award for Best PhD Thesis in 2017. In 2021, he was also honored with the Oral and Dental Research Institute Award. In 2026, he obtained The Excellence Award in Research in the Field of Medical Sciences and Its Technological Applications from Members' Club of Research. In 2025, he contributed, in collaboration with others, to a nationally registered patent. He was also recognized among the

top 2% of scientists worldwide according to the Stanford University ranking for the years 2024 and 2025.

Intelligent Dentistry: How Smart and 4D Materials Are Transforming Oral Care

Smart dental materials represent a transformative shift in modern dentistry, characterized by their inherent ability to sense environmental changes and respond accordingly. Also known as “responsive materials,” these systems can alter their properties in a controlled manner when exposed to stimuli such as stress, temperature, moisture, pH, or electric and magnetic fields. Recent advances have introduced a wide range of smart materials into the dental market, including smart resin composites, glass ionomer cements, pit and fissure sealants, endodontic instruments, orthodontic wires, silicone elastomers, bite registration materials, and thermochromic toothbrushes. Notably, chromatic dental composites exhibit adaptive color-matching capabilities, enhancing aesthetic outcomes. Furthermore, the emergence of 4D printing adds a dynamic dimension, enabling materials to change shape or function over time in response to environmental triggers. Collectively, these innovations promise to revolutionize clinical practice by improving treatment precision, durability, and patient-specific outcomes, paving the way toward a new era of intelligent and adaptive dental care.

Keywords: Smart, Biomimetic, bioresponsive, bio-smart dentistry, materials, 4D printing.

Ass.Prof. Eman Samy Shalaby

**Pharmaceutical Technology Department, Pharmaceutical and Industries Research
Institute, National Research Centre, Egypt**



Associate professor of pharmaceuticals published 19 scopus papers and a member of many national and international projects

Bilosomes as a promising approach for topical application of Allantoin intended against skin aging

Skin aging is one of the most common issues in the world, affecting nearly all women. Our study's objective was to evaluate allantoin bilosomes ability to prevent a rat model of aged skin. Allantoin bilosomes (ALL-BIL) were prepared and optimum formulations were selected that showed the highest encapsulation efficiencies, optimum particle size, and zeta potential. To improve application and industrial sustainability, the chosen formulation was included into a cream. The ALL-BIL loaded cream (ALL-BIL-C) showed pseudoplastic flow

and shear thinning behaviour. The effectiveness of ALL loaded chosen formulation for aging healing was evaluated which is reflected in the histopathological study. These results support the ALL-BIL-C's further development and assessment as a viable dermatological or cosmeceutical intervention and offer insights into the potential mechanisms by which it can exert its anti-skin aging benefits.

Dr. Salma Elmaaz

Master student at Founder, Solvira Pharma for Pharmaceutical Industries, Egypt



Salma Elmaaz is STEM student graduated senior 2021, Bachelor degree (B.Sc.) holder from Faculty of Science, Chemistry Department, Menoufia University with Excellent with honored. Master degree (M.Sc) organic chemistry student. She was selected as the top 20 scientists in the UNESCO Nanotechnology Program in Africa and studied a Nanotech diploma at the British University in Egypt. National Patent Holder from the Ministry of Scientific Research for inventing an ointment to treat burns naturally. Founder of Solvira startup creating the solutions to burn patients' challenges. Salma was a national finalist in Intel ISEF and won 1st place in the SCIENCE FAIR competition. UNFCCC YOUNGO Ocean's Voice team and was listed among the top 100 Arab youth and 1000 junior academy youths in New York. She won 1st place at ICSSD for best applicable product and ranked 1st in scientific research competitions at Menoufia University, GIZ Greener Hackthon and Nile University (UGRF) at Egypt level. Salma represented Egypt at COP27 and the RCOY MENA in Jordan. She won 3rd place in the Arab world at the Creative Student Forum in Jordan and was named Ideal Student of Egypt 2023. Salma graduated from the EcoSciGen Climate Fellowship and was selected as a top 150 Max Thabiso Edkins Climate Ambassador.

Synergistic Topical Cream Formulation for Burn Scar Reduction Using Achatinoidea Shell Powder

Burns are still one of the most important causes of death and disability in the world, where 800,000 people get burned per year. According to the World Health Organization, burns are the third-leading cause of death in the world. There are many ways to treat burns. One of these ways is by using a material called "betastestrol" that treats the burns without their effects. There are many ways to treat the effects caused by burns, such as lasers and surgical procedures, but these have negative effects like drying the skin. In this research, the project works to treat burns and eliminate their effects by using an ointment provided with nanotechnology to disinfect burns, treat infections, relieve pain, rejuvenate damaged cells, and repair body tissues; moisturize the skin and whiten it, give it freshness, remove burn-caused black spots, repair damaged tissues, regenerate damaged skin, and eliminate the effects of burns. The ointment treat burns and eliminate their deformities without using lasers or surgical procedures, and the treatment period is also very short; in the first degree, it takes

7 days to treat the burns and their effects. In the second degree, it takes 14 days. third degree, it takes about 22 days to treat the burns and their effects.

Ass.Prof. Mai Abd El Meguid Ali Dawoud Oada

Biotechnology Research Institute, National Research Centre, Egypt



Dr. Mai Abd El Meguid, PhD, is an Associate Professor of Microbial Biotechnology at the Biotechnology Research Institute, National Research Centre (NRC), Dokki, Giza, Egypt—one of the region's leading multidisciplinary research institutions. She earned both her MSc and PhD in biological sciences from the Faculty of Science, Ain Shams University, Cairo, Egypt. Dr. El Meguid is a molecular biologist whose research focuses on viral hepatitis and liver diseases, particularly infections caused by Hepatitis C Virus (HCV), Hepatitis B Virus (HBV), and Cytomegalovirus (CMV), which represent major public health challenges in Egypt and worldwide. Her work integrates molecular biology techniques with clinical relevance to advance understanding of viral pathogenesis, host immune responses, and the genetic and epigenetic factors involved in disease progression and hepatocellular carcinoma (HCC). Her long-term research goals include the development of early diagnostic biomarkers and preventive strategies for liver diseases. Throughout her career, Dr. El Meguid has authored more than 25 peer-reviewed research articles and reviews, primarily as first or corresponding author, published in respected international journals. Her work has received over 250 citations, with an h-index of 9 according to Google Scholar. In recognition of her academic excellence, she received the National Research Centre's Best MSc Award in 2012. Dr. El Meguid is an active member of several scientific societies, including the Egyptian Society for the Study of Gastroenterology and Liver Disease and the Organization for Women in Science for the Developing World (OWSD), and is a postdoctoral member of the American Society for Microbiology. She also serves as a peer reviewer and editorial board member for multiple international journals. Committed to mentorship and capacity building, she actively supports young researchers while fostering national and international scientific collaborations.

Evaluation of PNPLA3 genetic variant in Egyptian HBV-infected patients concomitant with metabolic-associated steatotic liver disease (MASLD)

Hepatitis B virus (HBV) infection and metabolic-associated steatotic liver disease (MASLD) increasingly coexist, creating complex diagnostic and therapeutic challenges. This study examined the association between the PNPLA3 rs738409 genetic variants and the progression of hepatic steatosis and fibrosis in HBV patients who have and do not have MASLD. A total of 170 Egyptian HBV patients were enrolled and categorized into two groups: non-MASLD

(n = 99) and MASLD (n = 71). The assessment of liver stiffness and hepatic steatosis was conducted using transient elastography (FibroScan®) and controlled attenuation parameter (CAP). Genotyping for PNPLA3 was performed via real-time PCR. The PNPLA3 GG genotype and G allele were significantly linked to the progression of hepatic steatosis in HBV-MASLD patients. Moreover, the frequency of CG+GG genotypes was higher in HBV-MASLD patients during both early (53.1% vs. 34.2%, $p=0.035$) and late fibrosis stages (73.7% vs. 33.3%, $p=0.014$) compared to non-MASLD patients. No association was found between PNPLA3 genotypes and HBV DNA levels. HBV viral load was notably associated with different fibrosis stages but was not linked to steatosis. HBV patients carrying the PNPLA3 GG genotype or G allele are more susceptible to developing MASLD and severe steatosis. Additionally, PNPLA3 variants may be implicated in the progression of fibrosis within the context of metabolic dysfunction; this association is not apparent in non-MASLD cases. Furthermore, viral dynamics appear to influence the progression of hepatic fibrosis but not steatosis. The concomitant MASLD in HBV patients was not associated with the severity of hepatic fibrosis.

Yousra Aly Ibrahim Sabry

Associate professor at Restorative and Dental materials Department, NRC



My name is Yousra Aly Ibrahim. I am an Associate Professor of Endodontics, Restorative and Dental Materials Department, Oral and Dental Research Institute, National Research Centre, Cairo, Egypt. I am an Associate Professor in the field of endodontics, passionate about combining clinical practice with research innovation; with particular interests in regenerative endodontics and novel therapeutic strategies for hard tissue repair. I was a member in two Dental Projects Team at the National Research Centre in Egypt from (2016-2019) and from (2019-2022).

Innovative Trends in Dentistry: A Shift Toward Conservative Concept

Modern healthcare is moving toward treatments that preserve natural tissues whenever possible, and dentistry is following the same direction. One of the most promising innovations is adult pulpotomy, a minimally invasive procedure that aims to preserve the healthy part of the dental pulp instead of removing it completely through conventional root canal treatment. Recent advances in bioactive materials have significantly improved the success of this approach. These materials not only seal the treated cavity; but also stimulate natural healing

and tissue repair, allowing the tooth to maintain its vitality and function. As a result, adult pulpotomy is becoming a reliable treatment option for carefully selected cases that were previously managed with more invasive procedures. This presentation highlights the scientific principles behind adult pulpotomy, the role of innovative biomaterials in its success, and its potential impact on the future of patient-centered dental care. It also discusses how this shift reflects a broader trend in healthcare toward regenerative, conservative, and biologically based treatment strategies.

Key words: adult pulpotomy, bioactive, healing, regenerative

Ass.Prof. Salma Ahmed El Marasy

Department of Pharmacology, Medical Research and Clinical Studies Institute, National Research Centre, Egypt



Associate Professor, Pharmacology Department, Medical Research Institute and Clinical Studies, National Research Centre (NRC), Egypt. PhD in 2015, in pharmaceutical sciences (Pharmacology and Toxicology), Faculty of Pharmacy, Cairo university. Published more than 25 articles related to experimental pharmacology in national and international journals. Participated as oral and Short talk (3 minutes) presenter in national and international conferences. Trainer in NRC Summer training courses for undergraduates 7 years from today's date. Coordinator and speaker in the workshops entitled "Ethics in Research" and "Tips and tricks in oral presentation skills" held at NRC, 2023. Coordinator and speaker of workshop entitled "Tips for Research Networking, writing an Impactful Research Paper and Grant Proposal" held at, 2022. Completed Nature Research Academics workshop entitled "Grant Writing", "Research Methodology" and "Writing a Research Paper" funded by STDF, 2022. Member in NRC Capital incubator funded by ASRT from March 2023 till now. Principle investigator of the project entitled: "Diabetes mellitus associated complications: Role of Advanced Nano-technology in the Delivery of Natural Flavonoids", NRC, 2020-2022. Principle investigator and workshop co-ordinator of "Researcher Connect project" under Newton Mosharafa fund, 2017 and under STDF fund, 2020 delivered by the British council at the NRC. Participated as facilitator in "Responsible conduct of research (RCR)" workshop funded by National Academy of Sciences-USA (NAS) and hosted by NRC on 2015 and 2016. Co-ordinator of "Researcher's Required Skills" workshop held at the NRC, 2016 and 2017. Co-ordinator and speaker in various symposiums and workshops held at the premises of the NRC in 2016. Finalist in Famelab Egypt 2015, attended Masterclass presented by science communicator/ media trainer, Mr. Karl Byrne in science communication to improve presentation and communication skills, won the audience vote. Attendant and participated in various modules held within the DAAD Kairo Academie. Participated as organizer in the 1st Scientific forum for applied medical research marketing, National Research Centre "The Future is Now" held at Nile Ritz Carlton Cairo-Egypt on December 17, 2016. Obtained certificate of appreciation as organizer in the 12th Annual congress of

Medical Sciences, under the theme ‘‘Medical breakthrough technologies towards a healthy community’’ within the National Research Centre in 2014.

**Anti-osteoporotic effect of Sitagliptin in an Osteoporosis Model of Ovariectomized Rats:
Role of RUNX2 and RANKL/OPG Ratio**

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This study examines the potential anti-osteoporotic effect of sitagliptin in osteoporosis instigated by ovariectomy (OVX) in rats. Rats were assigned into 4 groups: Sham-operated, OVX group, and OVX rats orally treated with sitagliptin (10, 20 mg/kg), respectively, after 8 weeks of OVX for 4 weeks. Biochemical, real-time polymerase chain reaction, histopathological, and immunohistochemical analyses of bone resorption and formation were conducted. Sitagliptin ameliorated bone mineral density (BMD), restored calcium and phosphorus levels in OVX rats, elevated catalase and decreased malondialdehyde, reduced receptor activator of NF- κ B ligand (RANKL), elevated osteoprotegerin (OPG), and reduced tartrate-resistant acid phosphatase (TRAP) femur contents. Sitagliptin mitigated variations in mRNA expressions of RUNX2 and protein kinase B (AKT) in femur tissue. Moreover, sitagliptin reduced caspase-3 protein expression and improved bone histomorphology and mechanical properties. Sitagliptin's anti-oxidant activity mediated its anti-osteoporotic effect in OVX rats via modulation of RUNX2, downregulation of RANKL/OPG, AKT pathways, apoptosis, and histomorphometry alterations revealing attenuation of osteoclastogenesis and promotion of osteoblast formation.

Toka A. Ahmed

Researcher, Egypt Center for Research and Regenerative Medicine



Dr. Toka A. Ahmed is an accomplished researcher with 11 years of experience in stem cell biology, tissue engineering, and regenerative medicine. She holds a Ph.D. in Biochemistry from Cairo University (2024), along with an M.Sc. (2019) and B.Sc. (2014) from Helwan University. Her expertise spans stem cell reprogramming, differentiation, 3D spheroid models, and biomimetic vascularized scaffold systems (e.g., amniotic membrane, Matrigel). Notably, her M.Sc. pioneered generating human iPSCs from adipose-derived pericytes, while her Ph.D. unravelled pericyte-diabetes crosstalk driving vascular complications. Dr. Ahmed led an STDF-funded project developing 3D lung spheroid models using custom microwell plates, yielding high-impact publications. Her work differentiating stem cells into neural-like cells directly supports brain organoid research. A proven grant winner and recipient of national and international awards, Dr. Ahmed frequently publishes in leading journals and serves as an ISSCR member and Q1 journal reviewer. Complementing her research is a 9-year teaching record in stem cell biology and chemistry, extensive mentorship of undergraduate and graduate students, and an IBCT international trainer certification.

Human Adipose-Derived Pericytes in the Pathophysiology of Type 2 Diabetes Mellitus

Pericytes (PCs) are multipotent perivascular cells that wrap around endothelial cells (ECs) to regulate blood flow and maintain vascular integrity. Type 2 diabetes mellitus (T2DM) induces PC detachment and dysfunction, contributing significantly to diabetic vascular complications. This study investigated the deleterious effects of the diabetic microenvironment on the regenerative capacities of human adipose tissue-derived PCs. PCs were cultured in diabetic serum (DS-PCs) or normal serum (NS-PCs) for 6, 14, and 30 days. Morphological analysis revealed that DS-PCs exhibited increased aggregation, distinct topographical invaginations, fragmented mitochondria, and thickened nuclear membranes. Functionally, DS impaired angiogenic differentiation, marked by compromised tube formation, downregulated VEGF-A, IGF-1, and COL21A1, and upregulated anti-angiogenic factors (TSP1, PF4, and the actin-related protein 2/3 complex). DS-PCs also demonstrated heightened apoptosis, evidenced by elevated expression of Clic4, Bcl-2-like protein, serine/threonine protein phosphatase, and caspase-7. Genomic and inflammatory profiling revealed dysregulated DNA repair genes (CDKN1A, SIRT1, XRCC5, TERF2), upregulated pro-inflammatory markers (ICAM1, IL-6, TNF- α), and altered expression of focal adhesion proteins (TSP1, TGF- β 2, fibronectin, PCDH7). Interestingly, despite these microenvironmental insults, DS-PCs preserved baseline intracellular reactive oxygen species (ROS) levels and upregulated key extracellular matrix (ECM) organizing proteins (vinculin, IQGAP1, and tubulin beta chain) as an adaptive resistance mechanism. Overall, while a diabetic microenvironment impairs the regenerative potential of PCs, their intrinsic cytoprotective mechanisms underscore their potential as a viable cell therapy.

Dr. Ragaa Fikry Fathy,

Veterinary Research Institute, National Research Centre, Egypt



Dr. Ragaa Fikry Fathy, a researcher at the Veterinary Research Institute, National Research Centre (NRC) in Dokki, Egypt, holds a PhD in biochemistry and biotechnology from Cairo University and is a Fulbright Postdoctoral Program nominee. Her academic research focuses on the impact of environmental pollutants on aquatic teleosts, aiming to develop sustainable solutions to mitigate such effects. Beyond her core research, she has contributed to numerous national and international conferences as both a speaker and an attendee, with her work published in prestigious Q1 and Q2 scientific journals.

Ecotoxicological Effects of Emerging Pharmaceuticals on African Catfish: Biomarkers for Neurotoxicity, Immunotoxicity, and Histopathology

Pharmaceutical residues have emerged as critical emerging contaminants, posing substantial threats to aquatic biodiversity and sustainable aquaculture. However, their sublethal effects on nontarget aquatic species remain underexplored. This study evaluates the comparative toxicity of three widely detected pharmaceuticals: gabapentin (GPT, anticonvulsant), valsartan (VAL, antihypertensive), and codeine (COD, opioid analgesic) on African catfish (*Clarias gariepinus*), a species of ecological and aquacultural importance. Fish were exposed for 15 days to environmentally relevant concentrations of GPT (79.86 $\mu\text{g/L}$), VAL (28.22 $\mu\text{g/L}$), and COD (5.27 $\mu\text{g/L}$). Neurotoxicity was assessed via acetylcholinesterase (AChE), monoamine oxidase (MAO), and cortisol levels; immunotoxicity was evaluated through lysozyme (LZM) activity, immunoglobulin M (IgM), and phagocytic activity (PA), while histopathological and histochemical changes were examined in the kidney, intestine, and spleen. All pharmaceutical exposures induced significant neuroendocrine and immune disruptions. AChE activity was reduced, while MAO and cortisol levels increased, indicating neurotoxic stress. Immunosuppression was evidenced by decreased LZM, IgM, and PA. Histological examination revealed tissue degeneration, mucosal damage, and lymphoid depletion, with PAS staining showing glycogen depletion across all treated groups. Among the tested compounds, VAL induced the most severe multisystemic effects, followed by GPT and COD. These findings highlight the overlooked ecological risks of pharmaceutical pollutants, underlining their potential to compromise fish health, disrupt aquaculture productivity, and threaten aquatic biodiversity even at low environmental concentrations. Such environmental challenges encourage the exploration of sustainable and environmentally safe methods for the disposal of these substances.

Radwa Hassan El-Akad

National Research Center



She has authored over 20 peer-reviewed publications in reputable indexed journals, with an h-index of 7 and her contributions extend to participating in international collaborations. She presented several workshops and hands-on training sessions focusing on practical approaches in metabolomics for researchers and post-graduate students. She continues to play a pivotal role in advancing metabolomics applications in natural product research.

Marine Metabolomics for Bioactive Compound Discovery in Sea Cucumbers"

Sea cucumbers are marine invertebrates with substantial bioactive compounds and considered food resource with high nutritional value. Metabolomic approaches enabled chemical profiling of two species of sea cucumbers collected in 2 different seasons and assigning biomarkers for each sample. In vitro cytotoxic activity assays against prostate and colon cancer cell lines showed its potent activity (IC₅₀ 6-12 microgram/millilitre) that was attributed to their triterpene content as revealed by multivariate data analyses. Researcher at the Pharmacognosy Department, National Research Centre, Egypt. She earned her M.Sc. and Ph.D. in Pharmacognosy from Faculty of Pharmacy, Cairo University. With 10-years of experience, Dr. El-Akad has specialized in the metabolomic profiling of natural resources applying advanced analytical tools such as UPLC/MS/MS and GC/MS which contributes to understanding the chemical diversity and therapeutic potential of natural resources.

Short talks

Dr. Naglaa Ramadan Atta Mohamed

Researcher, Cairo University Faculty of science, Egypt



Naglaa Ramadan Atta is a Master's graduate in Microbiology from the Faculty of Science, Cairo University, with over five years of experience in medical and food microbiology laboratories. has strong research experience, including experimental design, microbiological

analysis, and scientific writing, and has contributed to peer-reviewed publications derived from master's research. My research interests include microbiology, environmental sustainability, and the application of scientific innovation to address public health and environmental challenges.

Exploring the Therapeutic Potential of Postbiotics: Antimicrobial and Anticancer Properties from Sheep and Goat Sources.

Postbiotics, the metabolic byproducts of probiotic microorganisms, have gained attention due to their high and consistent bioactivity, making them suitable for pharmaceutical and nutraceutical development. This study investigates the postbiotic potential of mixed *Lactobacillus* strains isolated from the microbiota of goats and sheep, focusing on their antibacterial and anticancer properties. *Lactobacillus* strains were isolated through subculturing on MRS agar under anaerobic conditions. Probiotic potential was screened using Gram staining, catalase tests, and multiplex PCR for species identification. Postbiotics were extracted via sonication. Antimicrobial activity was assessed using the well diffusion method against key pathogens, and efficacy was measured through inhibition zones, Minimum Inhibitory Concentrations (MICs), and Minimum Bactericidal Concentrations (MBCs). Cytotoxicity was tested using the Sulforhodamine B (SRB) colorimetric assay, and IC_{50} values were determined for potent extracts. Postbiotics derived from goat and sheep microbiota exhibited significant antibacterial and anticancer activity. All formulations showed inhibitory effects against *Staphylococcus aureus*, *Streptococcus epidermidis*, *Escherichia coli*, and *Klebsiella pneumoniae*, with MIC values as low as 8 µg/mL. The formulations from goat buccal swabs (Postbiotic 2) and sheep nasal swabs (Postbiotic 3) were most effective. All postbiotics induced dose-dependent cytotoxicity against MCF-7 and HT-29 cancer cell lines, with minimal toxicity toward normal fibroblasts. Postbiotics from mucosal sources demonstrated superior bioactivity compared to fecal sources. These findings highlight the potential of postbiotics as safe and stable agents with dual antimicrobial and anticancer properties, supporting their development for pharmaceutical and nutraceutical applications.

Gehad kamel AbdalAl

Faculty of Nursing, Minia University, Egypt



My name is Jihad Kamel Abdelal, and I am a fourth-year undergraduate student at the Faculty of Nursing, Minia University, Egypt. I began my studies in nursing at the age of fifteen at a nursing school, where I completed the diploma program. After finishing the diploma, I pursued further studies at the nursing institute before joining the Faculty of Nursing, maintaining excellence in all semesters and achieving a distinction in every academic term.

Throughout my academic journey, I have developed a strong interest in scientific research, particularly in regenerative medicine and stem cell therapy, due to their potential to significantly improve patient care and healthcare outcomes. I have participated in previous research projects organized by my faculty, which provided me with valuable experience in study design, data collection, analysis, and interacting with research participants. My current research focuses on exploring the knowledge and attitudes of health sector students toward stem cells, aiming to understand their awareness, engagement, and readiness to adopt emerging therapies in clinical practice.

Exploring Future Healers' Views: Knowledge and Attitudes toward Stem Cells among Minia University Health Sector Students

This study aimed to investigate the depth of understanding and the attitudinal landscape surrounding stem cell science and its clinical relevance among medical and dental students at Minia University. It further sought to explore the interplay between knowledge levels and attitudinal dispositions, offering insights into future healthcare professionals' readiness to engage with regenerative medicine. Stem cells are unique cells capable of self-renewal and differentiation into specific cell types. They are classified as embryonic stem cells, which are pluripotent and can develop into any cell type in the human body, and adult stem cells, which have a limited differentiation potential restricted to their tissue of origin. Stem cells play a vital role in tissue maintenance, repair, and regeneration, and have therapeutic potential in treating various diseases, including blood cancers and neurological disorders. Stem cell research also raises ethical and legal considerations in some countries. This cross-sectional study aimed to assess the knowledge and attitudes of undergraduate students in medical and allied health faculties at Minia University regarding stem cells, their clinical applications, and ethical implications. Data were collected over four months from students aged 18 years and above across Medicine, Nursing, Pharmacy, Dentistry, and Allied Health faculties. A validated structured questionnaire was distributed offline and online via social media platforms, and incomplete responses were excluded. The questionnaire covered demographics, medical information, knowledge predictors, and attitudes. Data were analyzed using descriptive statistics.

Among 361 participants, most were female, under 21 years, first-year students, and from the Faculty of Nursing. The majority were aware of Egypt's stem cell donation programs and correctly identified key characteristics of embryonic and adult stem cells. Participants also demonstrated positive attitudes toward stem cell therapy and research. "Yes" and "Agree" responses were most frequent, while "No" and "Disagree" were least reported.

These findings indicate that health sector students at Minia University possess good knowledge and positive attitudes toward stem cells, reflecting readiness to engage with regenerative medicine in their future professional practice.

Amr Abdelsattar gamal

Faculty of nursing elminia university



Amr Abdelsattar Gamal is a fourth year nursing student at the Faculty of Nursing, Minia University. He has a strong academic interest in clinical nursing skills, patient safety, and evidence based nursing practice. Throughout his undergraduate studies, he has developed a solid foundation in nursing sciences, research principles, and clinical assessment skills essential for professional nursing practice. During his academic training, Amr participated in clinical placements across different hospital departments, including medical, surgical, and community health settings. These experiences enhanced his competencies in patient assessment, vital signs monitoring, infection prevention, therapeutic communication, and teamwork within multidisciplinary healthcare environments. His clinical exposure increased his awareness of the importance of accuracy, professionalism, and adherence to standardized clinical guidelines. Amr has a particular interest in improving the quality of nursing education and evaluating clinical performance among nursing students. He is involved in academic research focusing on knowledge and skills acquisition in fundamental nursing procedures, especially accurate blood pressure measurement. His research interests aim to bridge the gap between theoretical knowledge and safe clinical practice. In addition to his academic and clinical activities, Amr is interested in health promotion, continuous professional development, and the application of research findings to improve patient outcomes. He aspires to pursue postgraduate studies in nursing and contribute to nursing education, clinical research, and healthcare quality improvement. His long term goal is to support evidence based practice and enhance patient care standards within healthcare institutions through commitment, ethical practice, lifelong learning, and professional responsibility.

Evaluating the accuracy of blood pressure measurement knowledge and skills among nursing students

Accurate blood pressure measurement is a fundamental clinical skill in nursing practice and plays a critical role in diagnosis, treatment decisions, and patient safety. Inaccurate measurement may lead to misdiagnosis and inappropriate management of various health conditions. Therefore, ensuring that nursing students possess adequate knowledge and practical skills in blood pressure measurement before entering clinical practice is essential. This study aimed to evaluate the knowledge and skills related to blood pressure measurement among nursing students at the Faculty of Nursing, Minia University, and to assess the accuracy of their performance during clinical practice.

A descriptive cross-sectional observational design was utilized. The study was conducted at the Faculty of Nursing, Minia University, and included first-, second-, third-, and fourth-year nursing students who agreed to participate. Data were collected using two tools: a practical observational checklist and a theoretical assessment tool. The practical assessment evaluated students' performance during blood pressure measurement, including patient preparation, positioning, cuff selection and placement, palpatory and auscultatory techniques, timing, and accurate reporting of results. The theoretical assessment consisted of a 20-item test and picture-based questions developed according to the American Heart Association recommendations. To reduce potential bias, the practical assessment was conducted prior to the theoretical evaluation.

A pilot study was conducted on 10% of the sample to evaluate the clarity and applicability of the developed tools. Content validity was assessed by a panel of five nursing experts, and reliability was determined using Cronbach's alpha coefficient, which demonstrated acceptable internal consistency. The findings of this study are expected to provide valuable insights into nursing students' competencies and to support the development of targeted educational strategies aimed at improving blood pressure measurement accuracy and enhancing clinical practice outcomes.

Aya Dany

Faculty of Nursing, Minia University



Aya Dany is an undergraduate student at the Faculty of Nursing, Minia University. She graduated from the Technical Institute of Nursing with an "Excellent" grade, reflecting her academic dedication. Aya is deeply passionate about scientific research, with a particular focus on women's health and reproductive issues. Through her work, she aims to contribute to the field of nursing research by addressing critical health challenges, such as Polycystic Ovary Syndrome (PCOS). Her goal is to bridge the gap between clinical excellence and evidence-based practice to improve health outcomes for women in her community.

Bridging the Knowledge Gap: PCOS Awareness Among Female Students at Minia University

Background: Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder starting in adolescence, characterized by hormonal imbalances and ovarian dysfunction. Despite its impact, significant challenges persist in diagnosis due to gaps in awareness and health literacy among young women.

Objectives: This study aims to assess the awareness of Minia University students regarding PCOS symptoms, causes, and complications, while identifying primary information sources and misconceptions.

Methods: A cross-sectional study was conducted among 395 female undergraduate students at Minia University. The participants' age distribution was 67.8% (20-25 years), 31.4% (18-20 years), and 0.8% (above 25 years). The sample included students from medical faculties (55.3%) and non-medical faculties (44.7%), using a structured questionnaire administered electronically and in hardcopy.

Results: Preliminary findings revealed that 53% of the participants were unaware of PCOS, while only 47% had previous knowledge. Statistical analysis is currently underway to further explore the correlation between academic background and specific knowledge levels.

Conclusion: The results highlight a significant gap in PCOS awareness. This underscores the urgent need for targeted educational programs and health campaigns, especially in non-medical faculties, to ensure early detection and better health outcomes.

Basmalaa Ahmed Thabet Hassan

**Kasim Amein language school, AWS AI Scholar , IBM AI Engineering Specialization
certified, Egypt**



Basmalaa Ahmed Thabet Hassan is an Egyptian Biomedical Engineering researcher specializing in artificial intelligence for healthcare, computational medicine, and digital twin technologies. Her work focuses on developing AI-driven solutions for complex medical challenges, particularly in precision oncology and personalized healthcare. She is the founder of the Egyptian Biomedical Engineering Research Initiative (EBERI), a student-led platform promoting biomedical innovation and research training. Her research interests include AI-based disease modeling, computational neuroscience, and emerging healthcare technologies. Basmalaa has developed the GBM-Twin framework, an in-silico digital twin approach for glioblastoma research, integrating artificial intelligence, biomedical data analysis, and computational modeling to support personalized cancer research and clinical decision-making. Her work has been presented through international scientific forums and is progressing toward intellectual property protection. She has received recognition through scientific programs, innovation initiatives, and research opportunities, and actively contributes to STEM education and youth research empowerment. Through her interdisciplinary work, she aims to bridge artificial intelligence and medicine to advance the future of personalized healthcare.

An In-Silico Computational Simulation Framework for Glioblastoma Progression Modeling: Integrating 3D Transformer Segmentation, Computational Radiogenomic Feature Fusion, Simulation-Based Treatment-Response Modeling and Phantom-Based In-Silico Geometric Optimization

Glioblastoma multiforme (GBM, WHO CNS5 grade 4) is the most lethal primary brain tumour and provides a demanding testbed for computational modeling. Existing computational workflows for GBM are typically (i) fragmented — segmentation, growth modelling, treatment simulation and risk stratification are developed and evaluated in isolation, allowing errors to propagate without statistical accounting; (ii) non-identifiable — the diffusion (D) and proliferation (p) parameters of the Fisher–Kolmogorov (FK) reaction–diffusion model are known to be structurally non-identifiable from standard MRI, yet are routinely reported as point estimates; (iii) static — most published computational tumor-progression pipelines do not implement bidirectional belief updating from new imaging within a retrospective sequential-Bayesian sense; and (iv) computationally unconstrained in the geometric-optimization layer, in which residual tumour volume is minimized without regard to brain organisation.

Methods.

We propose an In-Silico Computational Tumor Progression Model: a fully retrospective, computationally validated, non-clinical research framework integrating seven coupled computational components: (1) a hybrid nnU-Net (v2) + Swin-UNETR multi-task segmentation backbone with deep supervision and test-time augmentation; (2) computational radiogenomic feature fusion that explicitly combines a 256-d imaging deep-feature vector,

107 IBSI-compliant hand-crafted radiomics, and a 6-d molecular vector (IDH, MGMT, 1p/19q, TERT, ATRX, EGFR) via a gated cross-modal attention block; (3) an Identifiable Fisher–Kolmogorov growth model in which the original (D, ρ) parameterisation is replaced by the front-velocity reparameterisation $v = 2\sqrt{D\rho}$, and identifiability is certified by four convergent tools (profile likelihoods, FIM eigenvalue rank, Sobol' indices, and Simulation-Based Calibration); (4) a retrospective sequential Hamiltonian Monte Carlo (HMC/NUTS) update loop that ingests follow-up MRI to refresh the posterior over v in a Bayesian sense; (5) a simulation-based treatment-response module combining linear-quadratic (LQ) radiobiology and Hill-equation pharmacodynamics with a dedicated pseudo-progression (PsP) disambiguation sub-module derived from RANO 2.0 image-derived features; (6) a Fine–Gray competing-risks risk-stratification module with isotonic-regression calibration; and (7) a phantom-based in-silico geometric-optimization module on 20 synthetic phantoms using the HCP-MMP1 cortical atlas and HCP-1065 tract templates to compute eloquence-aware computational penalties.

Results.

On 3,378 retrospective public cases (BraTS2023/2024, UPenn-GBM, UCSF-PDGM, TCGA-GBM, IvyGAP), the hybrid backbone reaches Dice 0.921 / 0.889 / 0.847 (WT/TC/ET) with voxel-wise ECE = 0.018. The re-parameterised FK model passes a four-tool identifiability certificate; sequential HMC reduces 90-day volumetric MAE from 12.4% to 7.1% ($p < 0.001$). The PsP module lowers false-progression labelling from 17.2% to 6.4% versus a single-timepoint RANO baseline. Fine–Gray + isotonic calibration achieves Uno C-index 0.761 (vs DeepSurv 0.743; $\Delta = +0.018$, $p < 0.001$) with time-dependent Brier ≤ 0.18 . Atlas-aware in-silico geometric optimization reduces simulated functional-region encroachment by 78% versus a volume-only objective on synthetic phantoms.

Conclusions.

To our knowledge this is the first in-silico GBM simulation framework that (a) resolves FK identifiability with a formal certificate, (b) integrates gated radiogenomic fusion and a PsP disambiguation sub-module, and (c) replaces volume-only objectives with atlas-aware computational penalties — all within a strictly retrospective, hypothesis-generating, non-clinical scope.

Mohammad Ahmed Ismael Ali Teleba

Faculty of nursing / Minia University



Mohamed Ahmed Ismael is a final-year undergraduate nursing student at the Faculty of Nursing, Minia University, Egypt. He is currently pursuing his Bachelor's degree in Nursing, with a strong academic interest in community health nursing, mental health promotion, epidemiology, and evidence-based healthcare. Throughout his undergraduate education, Mohamed has participated in academic research and health education activities, with a particular focus on mental health awareness among university students. His current research explores mental health literacy and the factors influencing students' knowledge, attitudes, and help-seeking behaviors. He has also contributed to developing research proposals, survey instruments, and community-based health education initiatives. Beyond his academic work, Mohamed is actively involved in creating and sharing educational medical content for nursing students and the public through digital platforms. His interests include health promotion, preventive healthcare, and scientific communication, with the goal of improving public awareness and supporting better health outcomes. Mohamed is committed to continuous professional development through scientific research, conferences, and interdisciplinary collaboration. He aspires to pursue postgraduate studies and build a career in nursing research, community health, and healthcare education, contributing to the advancement of nursing practice and public health in Egypt and beyond.

Awareness of Mental Health Disorders among Students at Minia University, Egypt: A Cross-Sectional Study

Mental health disorders represent a growing public health concern among university students, affecting academic performance, social relationships, and overall well-being. Despite the increasing burden of mental illness, mental health literacy remains limited in many developing countries, including Egypt. This study aims to assess the level of mental health literacy among undergraduate students at Minia University and identify the demographic and academic factors associated with it. A descriptive cross-sectional design will be conducted among approximately 400 students selected using stratified random sampling from medical and non-medical faculties. Data will be collected using the Mental Health Literacy Scale (MHLS), a validated instrument assessing knowledge, recognition of mental disorders, help-seeking confidence, attitudes, and social willingness toward individuals with mental illness. Descriptive and inferential statistical analyses will be performed using SPSS version 26, with statistical significance set at $p < 0.05$. The findings are expected to identify knowledge gaps, misconceptions, and stigma related to mental health, providing evidence for developing targeted awareness and educational programs within the university. Enhancing mental health literacy may promote early recognition of mental disorders, improve attitudes toward affected individuals, and encourage timely help-seeking among university students.

Nada Hamada Elsayed

Biotechnology student, Ain shams university



I am an undergraduate Biotechnology student at the Faculty of Agriculture, Ain Shams University, with a strong academic standing and practical expertise in molecular biology, bioinformatics analysis, and genetic research methodologies. As a recipient of the Egypt Scientists Research Scholarship, I have undergone intensive scientific training focused on research methodology, scientific thinking, and skill development. Additionally, I have completed specialized practical training programs covering PCR, qPCR, SDS-PAGE, and sequence alignment tools (BLAST/NCBI) at prominent institutions such as the National Research Center and Ain Shams University. Passionate about bridging molecular science and real-world applications, I have engaged in collaborative research projects, demonstrating strong skills in teamwork, scientific visualization, and research communication.

Title

The 16S ribosomal RNA (rRNA) gene sequencing has emerged as a cornerstone molecular marker for profiling bacterial communities within complex microbial ecosystems. However, significant methodological challenges such as primer biases, a lack of standardized protocols, and reproducibility issues persist and constrain clinical and environmental translation. Here we show a comprehensive synthesis of current advancements, critically evaluating both technical bottlenecks and emerging computational solutions in microbiome profiling. Our literature review reveals that integrating long-read sequencing technologies, multi-omics frameworks, and artificial intelligence-driven platforms significantly enhances taxonomic resolution and data accuracy. Furthermore, standardized data pipelines are essential to overcome inter-study variability and ensure reliable comparative analyses. These findings advance the field by establishing a clear roadmap toward standardized, high-resolution microbiome profiling systems, ultimately bridging the gap between genomic innovations and clinical applications.

Sayed Abdelrazek Abdelhamid
Researcher, National Research Centre, Egypt



Researcher in Microbial Biotechnology with more than 12 years of experience in industrial microbiology, microbial biotechnology, green nanotechnology, and sustainable bioprocess engineering. My research focuses on developing microbial technologies for the sustainable conversion of agricultural and industrial residues into biofuels, industrial enzymes, and functional nanomaterials. Through multidisciplinary research, I integrate microbial biotechnology, process optimization, and green nanotechnology to develop environmentally sustainable solutions for renewable energy, agriculture, environmental remediation, and biomedical applications and supporting the Circular Bioeconomy and Egypt Vision 2030.

Ecofriendly silver nanoparticles biosynthesis from *Penicillium commune* NRC 2016-3 as antimicrobial agents for textile materials

The biological synthesis of silver nanoparticles (AgNPs) using fungi has received considerable attention, given that they are environmentally friendly, non-toxic, and inexpensive. Therefore, the current work investigates the biological synthesis of silver nanoparticles (AgNPs) using the extracellular filtrate of *Penicillium commune* NRC 20163. Physicochemical analyses confirmed the formation of AgNPs with a surface plasmon resonance peak at 420 nm. Transmission electron microscopy (TEM) and Scanning electron microscopy (SEM) revealed spherical particles ranging in size from 3.69 to 9.61 nm, with a zeta potential of -30.5 mV, indicating good stability. The biosynthesized AgNPs exhibited strong antibacterial and antifungal activity against various test microorganisms, with inhibition zones ranging from 30 to 33 mm for Gram-positive and 26 to 30 mm for Gram-negative bacteria. The minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) values were found to be 25 μ g/mL and 2030 μ g/mL, respectively. Furthermore, the AgNPs-treated fabrics revealed good antimicrobial properties and demonstrated their durability even after multiple washing cycles. This finding highlights the potential of green-synthesized AgNPs for creating durable antimicrobial textiles.

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Ikram Benmakhloufi is a PhD student in Computer Science from Algeria. Her research focuses on artificial intelligence, deep learning, medical image analysis, edge AI, and on-device machine learning. She is particularly interested in developing lightweight AI models for smartphone- and edge-based medical image screening to support accessible healthcare in resource-limited settings. Her work aims to bridge the gap between advanced AI technologies and real-world clinical applications through efficient and deployable solutions.

Toward Accessible AI Healthcare: A Roadmap for Offline-First, On-Device Deep Learning in Medical Image Screening

Artificial intelligence (AI) has demonstrated remarkable potential in medical image analysis; however, its adoption remains limited in many low-resource healthcare settings due to unreliable internet connectivity, limited computing infrastructure, concerns over data privacy, and restricted access to specialist expertise. Recent advances in lightweight deep learning have enabled AI models to execute directly on smartphones and other edge devices, offering a practical alternative to cloud-dependent solutions. This presentation proposes a roadmap for the design and deployment of offline-first, on-device deep learning systems for medical image screening. The roadmap is structured around five key components: selecting lightweight neural network architectures suitable for resource-constrained devices; applying model optimization techniques, including pruning and INT8 quantization, while preserving diagnostic performance; designing offline-first workflows with optional synchronization capabilities; evaluating systems under realistic deployment conditions using device-level performance metrics alongside predictive accuracy; and incorporating usability, privacy, and clinical safety considerations into application development. In addition, the presentation discusses current challenges, including dataset bias, model generalizability, performance variability across mobile hardware, and the need for rigorous external and clinical validation before real-world adoption. Rather than introducing a new diagnostic model, this work synthesizes recent advances and practical implementation strategies to support researchers developing accessible, privacy-preserving, and sustainable AI-assisted medical screening tools. By bridging advances in edge AI with healthcare accessibility, the proposed roadmap contributes to the development of equitable digital health solutions aligned with Sustainable Development Goal 3 (Good Health and Well-being).

Keywords: Edge AI; Lightweight Deep Learning; Medical Image Screening; Offline-First Systems; Mobile Health.

Session 6 Online-Biology & others

Prof. Jibin Sun

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Dr. Sun is an expert in industrial synthetic biology and systems biotechnology from the Chinese Academy of Sciences, with over three decades of academic research, industrial transformation and international cooperation experience. He earned his PhD in Biotechnology and Bioinformatics from the German Research Centre for Biotechnology and Technische Universität Braunschweig in Germany between 2000 and 2004. Over there, he continued his postdoctoral research until 2008, before returning to China to join Tianjin Institute of Industrial Biotechnology (TIB), CAS as a full professor.

From 2012 to 2023, Dr. Sun served as Deputy Director General of TIB. During this pivotal tenure, he made contributions to China's synthetic biology infrastructure: he spearheaded the establishment of the National Center of Technology Innovation for Synthetic Biology and the core high-end biofoundry platform at TIB, driving the institute into China's top 1% ranking for technology transfer. Currently, he holds multiple roles including Principal Investigator and Director of the Center of Systems Biology at TIB, alongside key leadership positions in China's national microbiology, biotechnology and bioindustry associations. Recently, he was elected as Chairperson of the Coordinating Council of COMSTAS.

Academically, Dr. Sun has published more than 150 peer-reviewed international papers and filed nearly 100 invention patents worldwide, many of them successfully licensed to industry. His achievements have won numerous prestigious provincial and national science and technology awards across China.

Hayat Daeia

Third-year Bachelor of Science student majoring in Medical Sciences at Dalhousie University in Halifax, Nova Scotia, Canada



Hayat Daeia is an upcoming third-year Bachelor of Science student majoring in Medical Sciences at Dalhousie University in Halifax, Nova Scotia, Canada. Her academic interests center on women's health and wellness, with a long-term goal of specializing in this area within clinical medicine. In her first year, she was selected to join the Dalhousie Integrated Science Program, a competitive, research-focused cohort. Through this program, she completed two scientific research papers, including one conducted during her winter-term placement at the Neurocognitive Imaging Lab (NCIL) at Dalhousie University. She continues to volunteer with the lab, contributing to ongoing research through data collection and by assisting with literature screening for a graduate student's review paper. Hayat has presented her findings at several + conferences. She gave presentations at the Brain Recovery Center's Research Day in Halifax, Nova Scotia, and the Canadian Undergraduate Conference on

Healthcare in Kingston, Ontario. Additionally, Hayat actively participates in student advocacy and leadership. In addition to holding executive positions in a number of university societies, she represents students from the Middle Eastern and North African (MENA) diaspora as the Medical Sciences Society's Interpersonal Diversity and Equity Representative. Hayat takes part in local environmental initiatives. She was awarded a grant to create and present composting workshops at local libraries that encourage composting and lessen food waste after taking part in multiple programs focused on girls' entrepreneurship and climate change.

Brain-computer interfaces (BCI)

Brain-computer interfaces (BCI) utilize machine learning classifiers to categorize specific brain signals into recognizable commands. These signals are typically recorded using electroencephalography (EEG) that measure various neural responses such as steady-state visually evoked potentials (SSVEPs). SSVEP signals are brain responses primarily generated in the visual cortex, where neural activity synchronizes with the frequency of continuously flickering visual stimuli. These signals are processed and classified using machine learning algorithms, with classification accuracy measured. A major challenge in SSVEP-BCIs is visual fatigue, prompting research to enhance signal strength and accuracy while maintaining user comfort. Previous studies on optimal stimulus parameters have yielded conflicting results, often involving trade-offs between stronger responses and user comfort. In this study, we investigate whether a novel aperiodic shape, the hat tile, can improve SSVEP signal to noise ratios (SNR) and classification accuracy compared to a square stimulus. We hypothesized no significant differences. We tested this by collecting data from seven participants (ages 18-56), comparing amplitudes of SSVEP responses, and evaluating accuracy. Our findings refute our hypothesis; the square yielded higher SNR and accuracy. Given limitations such as small sample size, electromagnetic interference, non-normally distributed data, and not testing a checkerboard-like stimuli, further research is warranted to explore the potential of aperiodic stimuli in SSVEP-BCI applications.

Dr Zohra Aloui-Zarrouk

Research Scientist at Institut Pasteur de Tunis (IPT) and ex-lecturer in biochemistry at Tunis ElManar University



Dr Zohra Aloui-Zarrouk, PhD in biochemistry, is a Research Scientist at Institut Pasteur de Tunis (IPT) and ex-lecturer in biochemistry at Tunis ElManar University. At the Molecular Epidemiology and Experimental Pathology Laboratory (MEEP) at IPT, she contributed and lead several projects mainly the structural and pharmacological characterization of novel

proteins from snake venom related to VEGF and the valorization of natural plant products for the development of novel intervention tools and strategies against infectious diseases mainly drugs for prevention or therapy.

Zohra is an African Academy of Sciences (AAS) Affiliate since 2017, and she lead the Working Group in “Basic and Applied Sciences” With the aim to strengthen more collaborations between African researchers.

With a particular interest in Scientific Entrepreneurship and the valorization of research outcomes, Zohra is a recipient of a PAQ-PAES project funded by the World Bank to create an Spin-off related to the R&D environment. This Spin-off is incubated at the Institut Pasteur of Tunis, Tunisia.

"Bridging Science and Entrepreneurship: The Emergence of Zora Pharm. Emoji as a Scientific Spin-off for innovative Plant-based Dermaceuticals.

Skin lesions represent a wide spectrum of pathological conditions and constitute a major clinical and socio-economic burden, particularly in low- and middle-income countries. These lesions are frequently associated with inflammation, infection, delayed healing, and permanent scarring, highlighting the need for therapeutic strategies that simultaneously control inflammation and promote tissue regeneration. Limited access to healthcare services and the high cost and side effects of conventional pharmaceuticals further exacerbate this burden. Medicinal and aromatic plants are a valuable yet underutilized source of bioactive compounds with demonstrated antioxidant, anti-inflammatory, anti-infectious, and regenerative properties. Africa's rich botanical diversity offers significant opportunities for the development of innovative and affordable therapeutic solutions; however, these resources remain insufficiently translated into standardized pharmaceutical and dermo-cosmetic products. Our Spin-off project focuses to develop and characterize novel skin-healing formulations based on natural active ingredients, including essential oils and plant extracts previously identified and studied in our laboratory. To enhance bioavailability, skin deposition, and therapeutic efficacy, these actives will be incorporated into advanced-based delivery systems, such as bio-polymeric carriers and lipid nanoparticles. Quality control and biological activity are evaluated using appropriate in vitro cellular models according to pharmacopeia and ISO standards. Our ultimate goal is to produce safe, effective, and affordable skin-care formulations derived from natural sources, with reduced reliance on harmful chemicals. By leveraging readily available raw materials, this project seeks to add value to medicinal and aromatic plants while addressing a critical healthcare need at both national and international levels.

Bishayr Hassan Aljaffar

King Faisal General Hospital



Bishayr Hassan Aljaffar is a dedicated Nursing Specialist from Saudi Arabia, recognized for her commitment to patient-centered care and professional excellence. Born on September 4, 1991, she has built her career on compassion, resilience, and a strong sense of responsibility toward improving healthcare outcomes.

Predictors of Exclusive Breastfeeding in Employed Mothers: An Ecological Approach-

This study examined the gendered predictors of exclusive breastfeeding among employed mothers in Saudi Arabia using Bronfenbrenner Ecological Systems Theory as a guiding framework. A cross-sectional design was employed, with a random sample of 201 employed mothers recruited from two hospitals and two primary healthcare centers in Alâ Ahsa. Data were collected through validated Arabic questionnaires addressing individual, microsystem, exosystem, and macrosystem factors. Hierarchical logistic regression was used to analyze predictors across ecological levels. Findings revealed that 36.8% of employed mothers did not practice exclusive breastfeeding at six months postpartum. The final regression model explained 42.9% of the variance in breastfeeding practices. Significant predictors included perception of milk supply (OR = 1.029), maternal mental health status (OR = 0.931), workplace support (OR = 1.024), and social norms (OR = 2.009). These results highlight the multi-layered influences shaping maternal health behaviors, where individual perceptions, psychological well-being, organizational policies, and societal expectations intersect to determine breastfeeding outcomes. The study underscores the importance of gender-sensitive interventions that address maternal confidence, mental health, workplace flexibility, and community attitudes. Healthcare providers should routinely assess mothers' perceptions of milk sufficiency and mental health, while advocating for supportive workplace policies. Public health campaigns targeting social norms are also essential to empower women in sustaining exclusive breastfeeding. In conclusion, exclusive breastfeeding among employed mothers is not merely a personal choice but a gendered health behavior shaped by ecological systems, requiring integrated strategies across clinical, organizational, and societal levels.

HAMEEDA MATOOQ ALJANABI

Dammam Health Network Eastern Health Cluster, Saudi Arabia



Hameeda Matooq Aljanabi is a highly accomplished Senior Nurse Educator and Nursing Professional Development Specialist, currently serving within the Dammam Health Network “ Eastern Health Cluster. With a strong academic foundation, she holds a Master of Science in Nursing and is pursuing her PhD, reflecting her dedication to advancing nursing education, leadership, and evidence ‘based practice. Her career spans extensive experience in clinical nursing, academic instruction, and professional development. As a Registered Nurse and Advanced Clinical Nurse Educator (ACNE), Hameeda has played a pivotal role in shaping nursing standards, mentoring staff, and implementing innovative educational strategies that enhance patient care and workforce readiness. She is recognized for her ability to integrate theory with practice, ensuring that nurses under her guidance are equipped with both technical competence and compassionate care values. Hameeda expertise includes curriculum design, competencyâ€‘based training, and the development of professional growth pathways for nurses across diverse healthcare settings. She has contributed to initiatives that strengthen nursing leadership, improve patient safety, and align clinical practice with international accreditation standards. Her work emphasizes continuous improvement, collaboration, and the empowerment of nurses to meet evolving healthcare challenges. As a PhD candidate, Hameeda is committed to advancing research in nursing education and professional development, with a vision of transforming healthcare delivery through knowledge, innovation, and leadership. Her career reflects excellence, resilience, and a passion for elevating the nursing profession in Saudi Arabia and beyond.

Knowledge and Attitudes Toward PICC Line Care Before and After Specialised Training in the Dammam Health Network, Saudi Arabia-

Peripherally Inserted Central Catheters (PICC) are widely used for long‘term intravenous therapy, yet their safe and effective management depends on nurses knowledge, attitudes, and adherence to evidenceâ€‘based practices. Inadequate training can result in serious complications, including infection, thrombosis, and catheter occlusion, highlighting the need for structured educational interventions. This study assessed the effectiveness of a specialized training program in improving nursesâ€™™ knowledge and attitudes toward PICC care within the Dammam Health Network (DHN), Kingdom of Saudi Arabia. A quasi‘experimental pre and post ‘test design was employed, involving 180 registered nurses who completed a two ‘week program consisting of six sessions integrating theoretical instruction and practical demonstrations. Knowledge and attitudes were measured before and after training using validated assessment tools. Results revealed statistically significant improvements ($p < 0.05$) across multiple domains. Notable gains were observed in understanding aseptic techniques (from 63.3% to 87.2%, $p = 0.001$), flushing protocols, infection prevention strategies, and documentation standards. In addition, participants reported increased confidence in their ability to manage PICCs and expressed readiness to engage in continuous professional development. The findings confirm that targeted training substantially enhances nurses

competencies in PICC care, reduces risks of catheter-related complications, and fosters a culture of safety and accountability. This study underscores the importance of regular, tailored educational programs as part of professional development initiatives, ensuring that nurses remain equipped to deliver high quality patient care. Integrating such training into institutional practice can strengthen clinical outcomes and advance nursing standards across healthcare systems.

Dr. Anwar Omar Dawod

University of Dohuk - College of Pharmacy, Iraq



I am Anwar Omar Dawod, a Pharmacist and researcher with a strong academic and clinical interest in biochemical monitoring and polypharmacy effects. I graduated from the University of Duhok, College of Pharmacy, in 2015. Since graduation, I have gained extensive experience working in hospital and community pharmacy settings, including clinical practice at Heevi Pediatric Hospital, Dohuk and other pharmaceutical centers. I am also involved in community pharmacy practice through my own pharmacy, Norsi Pharmacy. In addition, I have attended many scientific conferences including Pluxes (Croatia, 2024), Kurdistan Hematology Summit (2023) and others. I have participated with WHO (2016) in two periods in the implementation of ECAP in Duhok. I have participated in many pharmaceutical scientific and selling skills trainings including: BD phaSeal system (2023), training program of portfolio medical review and selling skills in 2018 and others. I am currently conducting a Master's degree research project entitled "Biochemical Monitoring of Hospitalized Elderly Patients Subjected to the Use of Polypharmacy in Duhok." My research focus on evaluating biochemical parameters such as liver function tests, renal markers, inflammatory markers, oxidative stress indicators, and drug-drug interactions among elderly patients exposed to multiple medications. My work integrates clinical pharmacy, laboratory biomarkers, and statistical analysis using SPSS to assess the impact of polypharmacy on patient health outcomes.

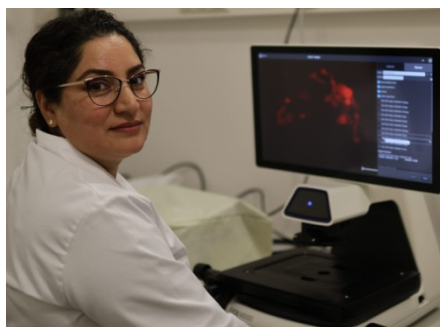
Monitoring hospitalized elderly patients and the use of polypharmacy in Dohuk: a biochemical evaluation

Aging and polypharmacy in hospitalized elderly patients are associated with physiological stress, decline in renal and hepatic function, oxidative damage, chronic inflammation, and increased susceptibility to drug-drug interactions. Objective: This study evaluated the biochemical effects of aging and polypharmacy on renal, hepatic, inflammatory, oxidative

stress, and nutritional biomarkers, while examining gender-specific differences and interaction frequency. Methods: A cross-sectional study enrolled 250 elderly patients admitted to Azadi Hospital, Duhok. Participants were classified into two age groups: 50–64 years and ≥ 65 years. Polypharmacy was defined as concurrent use of 5, 6–9, or ≥ 10 medications. Measured biochemical parameters included renal markers (urea, creatinine), hepatic enzymes (AST, ALT, ALP, bilirubin), oxidative stress marker malondialdehyde, inflammatory marker C-reactive protein, and nutritional indicators (albumin, hemoglobin). Drug–drug interactions were evaluated using the Medscape Drug Interaction Checker. Results: 30.8% of patients were receiving ≥ 10 medications. Increasing medication burden significantly elevated malondialdehyde levels (+84% to +139.8%), reduced albumin concentrations (–18.2% to –23.3%), and increased pancreatic amylase activity (+5.9%). Patients aged ≥ 65 years demonstrated renal impairment, particularly females, with elevated urea (+62.6%) and creatinine (+168.2%). C-reactive protein remained elevated (87.21 ± 81.17 mg/L), indicating systemic inflammation. Drug–drug interaction increased with medication number, with major interactions predominating among patients receiving ≥ 10 medications. Females showed broader age-related biochemical deterioration, whereas males demonstrated greater susceptibility to oxidative stress and hypoalbuminemia. Conclusion: Aging with polypharmacy worsens organ dysfunction and inflammation, emphasizing the need for individualized pharmacotherapy and routine biochemical monitoring to improve patient safety and clinical outcomes overall.

Dr. Amna Mehmood

Senior Scientist at the Institute of Microbiology, Martin Luther University Halle-Wittenberg, Germany



Dr. Amna Mehmood is an Afghan molecular biologist and Senior Scientist at the Institute of Microbiology, Martin Luther University Halle-Wittenberg, Germany. As a child, she fled Afghanistan with her family to Pakistan, where she completed her Bachelor's degree in Biotechnology. She later moved to Germany to pursue an M.Sc. in Molecular Genetics and subsequently earned her Ph.D. in Biochemistry at Jacobs University Bremen through a prestigious DAAD scholarship.

Before returning to academia, Dr. Mehmood served as a Senior Molecular Biology Specialist at the Ministry of Public Health, Afghanistan, where she worked on Global Fund and WHO projects supporting the establishment and strengthening of national reference laboratories for infectious disease diagnostics.

Since 2017, she has been conducting interdisciplinary research at Martin Luther University Halle-Wittenberg, initially as a Project Leader and currently as a Senior Scientist. Her research integrates cancer genomics, molecular imaging, protein engineering, and microbial biochemistry, with applications ranging from whole-exome sequencing of breast cancer and photoacoustic reporter proteins to the molecular mechanisms of [NiFe]-hydrogenase maturation.

Beyond her scientific work, Dr. Mehmood is a committed advocate for women in STEM, science diplomacy, and equitable access to education. Through international collaborations and community initiatives, she works to strengthen scientific capacity and expand opportunities for women and displaced scholars worldwide.

Clinico-Genomic Characterization of Breast Cancer in an Understudied Population: From Whole-Exome Sequencing to Precision Oncology

Breast cancer is the most frequently diagnosed cancer among women worldwide; however, many populations remain underrepresented in genomic research, limiting the implementation of precision medicine. Expanding genomic studies to diverse populations is essential for identifying population-specific genetic variants and improving personalized cancer care.

In this presentation, I will highlight our work on the clinico-genomic characterization of breast cancer in women from Khyber Pakhtunkhwa (KP), Pakistan, one of the first comprehensive whole-exome sequencing studies in this understudied population. We identified both somatic and germline variants in key DNA damage response genes, including **ATM, CHEK2, PALB2, and XRCC2**, with several recurrent mutations demonstrating potential as population-specific biomarkers. To investigate their biological significance, genomic analyses were complemented by structural bioinformatics, protein interaction network analysis, molecular docking, and molecular dynamics simulations, providing insights into the functional consequences of these variants and their potential implications for targeted therapy.

This work demonstrates how integrating genomics with computational structural biology can improve our understanding of cancer biology and contribute to more equitable precision oncology for underrepresented populations. The study was conducted as part of a **DAAD-supported international collaborative research project between Germany and Pakistan**, highlighting the importance of international scientific partnerships in strengthening research capacity and advancing global health research.

In addition to this work, I will briefly present how my research has expanded into molecular imaging through the development of photoacoustic reporter proteins, protein engineering, and the molecular mechanisms of [NiFe]-hydrogenase maturation in *Escherichia coli*. Together, these projects illustrate how interdisciplinary and collaborative research can bridge

fundamental molecular biology with translational biomedical applications while fostering scientific cooperation across borders.

Dr. Iman Abdulmanaf Taha

University of Dohuk, College of Science, Iraq



My name is Iman Abdulmanaf Taha, and I was born on 14 April 1999. I live in Kurdistan, Iraq. I'm currently a researcher studying a Master's at the University of Duhok, College of Science, Chemistry Department. The theme of my MSc study is "Evaluation of biochemical markers and anthropometric parameters in children diagnosed with phenylketonuria in Northern Iraq and Nineveh." Driven by a deep curiosity and passion for science, after earning my Bachelor's degree (2021-2022), I continued my academic journey with determination and a strong commitment to research and learning. In addition to my studies, I have been volunteering at the Hivi Pediatric Hospital for more than two years. This experience has strengthened my practical skills and enhanced my sense of responsibility, compassion, and dedication to serving others. Through both my academic and volunteer work, I have grown professionally and made a positive impact in the community. I have joined the International Conference on the Role of Universities in Achieving Sustainable Development Goals (ICRU-SDGs), Theme: "Advancing The SDGs Through Education, Research, and Innovation" held on the 23rd and 24th of September, 2025.

**Biochemical and Developmental Outcomes in Iraqi Children with Phenylketonuria
Following Late Diagnosis and Dietary Intervention.**

Phenylketonuria (PKU) is an autosomal recessive disorder requiring early detection via newborn screening (NBS) to prevent neurological impairment. In Northern Iraq, conflict has disrupted healthcare, leading to delayed diagnoses and inadequate management of PKU patients. Objectives: To evaluate the clinical and demographic profile and outcomes of children with PKU. Methods: a cohort of 19 children diagnosed with PKU in Kurdistan Region of Iraq (KRI) and Nineveh Governorate was evaluated through phenylalanine level quantification and application of questionnaires. Results: children's age ranged from 1 month to 15 years with male predominance (58%). Balanced ethnic distribution between Iraqi Kurdish non-Yazidi and Iraqi Kurdish-Yazidi (47% each) was identified. High consanguinity rate was found (84%) with 62.7% of marriages among first-cousins. 54% of the parents did not perform pre-marital medical exams. Moderate PKU appeared as the main phenotype (53%) followed by classic PKU (41%). Delayed diagnosis was found, with 53% having

received the PKU diagnosis at the age of 1-5 years., followed by the age range of 6-10 years (16%). High incidence of PKU misdiagnosis appeared (79%), where the most prevalent type of misdiagnosis was cerebral palsy (47%). The incidence of seizures (81%) was found associated with cerebral palsy (60%) and microcephaly (20%). Conclusions: the findings emphasize the need for implementation of comprehensive newborn screening programs, integration of metabolic testing into pediatric neurology evaluations, integration of genetic counseling into premarital services, besides the need for specialized metabolic professionals in the region. Strengthening these preventive and diagnostic strategies is essential to reduce PKU mortality.

Folasade M. Olajuyigbe

**Enzyme Biotechnology and Environmental Bioremediation Research Unit, Department
of Biochemistry, School of Life Sciences, Federal University of Technology, Akure,
Nigeria**



Professor Folasade Mayowa Olajuyigbe is an internationally recognized biochemist, structural biologist, and research leader whose work in enzyme biotechnology and environmental bioremediation is advancing sustainable solutions to global health and environmental challenges. She is Professor of Applied Biochemistry at the Federal University of Technology, Akure (FUTA), Nigeria, where she currently serves as Director of the University Central Research Laboratory. She is also the National Chairperson of the Organization for Women in Science for the Developing World (OWSD), Nigeria Chapter, and Project Coordinator of the United States Government Exchange Alumni Association of Nigeria (USGEAAN), Ondo State Chapter. Professor Olajuyigbe leads the Enzyme Biotechnology and Environmental Bioremediation Research Unit (E-Biotech Lab) as Principal Investigator. She has successfully supervised more than 50 postgraduate researchers, including 10 Ph.D. and over 40 M.Tech. graduates. Her multidisciplinary research integrates structural biology, enzyme biotechnology, microbial biocatalysis, bioenergy, and environmental bioremediation, with applications in structure-based drug discovery, industrial biotechnology, agricultural waste valorization, renewable energy production, and sustainable pollution control. Her research has contributed significantly to the development of enzyme-based technologies for biomass conversion and environmental remediation, resulting in over 50 peer-reviewed publications in reputable international journals and more than 1,100 citations. A Fulbright Senior Research Scholar at Rutgers University, USA (2017–2018), Professor Olajuyigbe has secured competitive research grants from the International Centre for Genetic Engineering and Biotechnology (ICGEB), Italy; Nigeria's Tertiary Education Trust Fund (TETFund) National Research Fund; and the International Foundation for Science (IFS), Sweden. She has also won international conference and travel grants from Nature, IFS, and the American Society for Biochemistry

and Molecular Biology (ASBMB). Her distinguished career has earned her numerous honours and awards, including two Schlumberger Foundation Faculty for the Future Fellowships, Associate Award of the International Centre for Theoretical Physics (ICTP), Italy, the 2024 Seeding Labs Instrumental Access Award for the FUTA Central Research Laboratory, and multiple FUTA Excellence and Productivity Awards under the Vice-Chancellor's Special Recognition category. Widely respected as a mentor and advocate for women in STEM, she has received national and international recognition for her leadership in advancing gender equity and scientific capacity development. Professor Olajuyigbe is an active member of several professional organizations, including the Nigerian Society of Biochemistry and Molecular Biology (NSBMB), the American Society for Biochemistry and Molecular Biology (ASBMB), the West African Research Association (WARA), and the American Association for the Advancement of Science (AAAS). She is a regular invited speaker at national and international scientific conferences, where she has delivered keynote, plenary, and lead papers. Outside academia, Professor Olajuyigbe enjoys creative cooking and window shopping. She is happily married to Revd. Canon (Dr.) Ebenezer Olajuyigbe, and they are blessed with children.

Enzyme Biotechnology: Catalyzing Sustainable Solutions for a Better Future

Folasade M. Olajuyigbe

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Enzyme biotechnology is transforming the global pursuit of sustainable development by replacing energy-intensive and environmentally hazardous processes with efficient, renewable, and eco-friendly biocatalytic systems. This paper highlights the pivotal role of enzyme biotechnology in driving the circular bioeconomy through innovative applications in biomass valorization, environmental remediation, and renewable energy production. Drawing on our research, we demonstrate how the discovery, characterization, and engineering of robust microbial enzymes have significantly enhanced the bioconversion of lignocellulosic agricultural residues into fermentable sugars for biofuel production. Particular emphasis is placed on lignocellulolytic enzymes, including cellulases, β -glucosidases, laccases, and other lignin-degrading oxidoreductases, as well as multifunctional enzyme systems that enable efficient delignification and saccharification of recalcitrant biomass. Our findings further demonstrate the remarkable potential of free and immobilized enzymes in the degradation of persistent pollutants, including synthetic dyes, bisphenol A, and petroleum hydrocarbons, thereby advancing sustainable waste management and ecosystem restoration. Furthermore, innovations in enzyme immobilization and recombinant DNA technology have substantially enhanced enzyme stability, catalytic performance, and industrial scalability, addressing key challenges associated with process efficiency and economic viability. The exploration of indigenous microbial biodiversity has also expanded the portfolio of industrially relevant biocatalysts adapted to diverse environmental conditions. Collectively, these advances establish enzyme biotechnology as a powerful catalyst for integrating waste valorization, pollution mitigation, and clean energy generation into sustainable bioprocesses. Continued investment in enzyme discovery, protein engineering, and bioprocess innovation will accelerate the development of resilient, cost-effective, and environmentally responsible technologies capable of supporting global sustainability goals and a thriving bio-based future.

Others

Dr. Noushin Arfatahery

Freie Universität Berlin, Germany Founder & Director, WISEA (Women in STEM & Environmental Advocacy) Germany



Dr. Noushin Arfatahery holds a Doctorate in Biology from Freie Universität Berlin, Germany. Her research focuses on microbial ecology, sustainable aquaculture, food microbiology, environmental microbiology, and women's health. She is currently collaborating with the Leibniz Institute of Freshwater Ecology and Inland Fisheries (IGB), Germany, and IRTA, Spain, on Integrated Multi-Trophic Aquaculture (IMTA) and microbial ecology research. Dr. Arfatahery is the Founder and Director of Women in STEM and Environmental Advocacy (WISEA), an initiative dedicated to advancing gender equality, scientific collaboration, and environmental sustainability. She serves as an American Society for Microbiology (ASM) Ambassador in Germany, a Reviewer for the UNESCO Science Report (8th Edition), and a UNESCO-TWAS Young Affiliate (2021–2026). Her current work integrates microbial ecology, artificial intelligence, and sustainable food systems to develop innovative solutions for aquaculture and global food security.

Empowering Women Scientists as Drivers of Inclusive Innovation and Sustainable Development

Gender equality is a critical enabler of scientific innovation and sustainable development, yet women scientists continue to face systemic barriers that limit their leadership, visibility, and participation in decision-making processes. Addressing these inequities is essential not only for social justice, but also for strengthening the quality, relevance, and impact of scientific research worldwide. This presentation draws on interdisciplinary research and leadership experience in environmental microbiology, sustainable aquaculture, and gender-inclusive scientific initiatives across Europe and the Global South. It highlights how empowering women scientists contributes to innovation-driven solutions in areas such as food security, environmental resilience, and public health, while fostering more inclusive and responsive research ecosystems. Through applied research projects and international collaborations integrating microbial ecology, circular food systems, and community engagement, the presentation demonstrates how women-led scientific initiatives can translate knowledge into socially relevant outcomes. Particular emphasis is placed on mentorship, capacity building, and cross-sector collaboration as mechanisms for enabling women researchers to move from scientific training to leadership and innovation roles. The contribution argues that gender

equality in science must be recognized as a strategic investment in sustainable development rather than a peripheral objective. Embedding gender-inclusive policies within research institutions, funding frameworks, and innovation ecosystems enhances scientific excellence and societal impact simultaneously. By positioning women scientists as key actors in innovation and entrepreneurship, science can more effectively respond to global challenges and contribute to resilient, equitable, and sustainable futures.